Low-Energy Neutrino Event Reconstruction: Clustering, Vertexing, and Energy Resolution in Phase I

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Abstract

The DUNE aims to measure the two most energetic components of the flux of solar neutrinos (^{8}B & *hep*), as well as neutrinos from other astrophysical sources such as nearby supernovae or the diffuse supernovae neutrino background. The sensitivity to the ν-Ar Charged Current (CC) channel, with orders of magnitude higher cross-section than the ν elastic scattering, together with DUNE's high exposure and great energy resolution, present an exceptional advantage in understanding the characteristics of these neutrinos. This work enables the reconstruction of low-energy neutrino events and demonstrates the energy resolution performance of the CC channel, under the presence of significant radiological and external backgrounds. Described are the development and validation of a **near-threshold** clustering algorithm that identifies primary ionization signals and associates adjacent energy depositions to account for spatially correlated activity. Additionally, the association of charge and light signals is described, from which the drift coordinate reconstruction can be performed.

Dedicated energy calibration strategies are employed to recover the interacting neutrino energy. The impact of these on the various background scenarios is assessed by reporting on the resolution achieved in each Far Detector (FD) drift volume. Highlighted are the algorithm's efficiency in reconstructing low-energy neutrino interactions ($<$10 MeV), the accuracy of the reconstructed vertex position ($<$1 cm), and the energy resolution ($<$20 %). The results are shown to be stable across the different detector geometries and background conditions. This demonstrates the robustness of the reconstruction approach and its potential contribution to the solar-ν physics program in DUNE.

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1 Introduction

The DUNE is a next-generation long-baseline neutrino detector. With its four large 14 kt Far Detector (FD) modules [1], it will be sensitive to the two most energetic components of the solar neutrino flux (^{8}B and *hep* neutrinos), as well as, neutrinos from nearby occurring supernovae [2]. The high cross-section of Charged Current (CC) interaction on argon nuclei, together with DUNE's large active mass and excellent energy resolution, will allow for precision measurements of these astrophysical neutrino fluxes and their energy spectra. These measurements will provide deep insights into neutrino properties (the MSW effect), solar physics (the Sun's metallicity and core temperature), and potentially new physics beyond the Standard Model (non-standard interactions, sterile neutrinos, etc.).

However, the sensitivity to low-energy neutrinos in DUNE is challenged by significant radiological and external backgrounds, which will smear the signals and generate false event candidates reducing the overall sensitivity. Developing robust reconstruction techniques that can efficiently identify and reconstruct low-energy neutrino events while suppressing backgrounds is crucial for realizing DUNE's full potential in the low-energy neutrino range (5 MeV to 100 MeV).

In this work, we present a comprehensive study of low-energy neutrino event reconstruction in DUNE, focusing on clustering, vertexing, and energy reconstruction algorithms. Evaluated is the performance of these techniques for the different FD background scenarios, demonstrating their robustness for enabling precision measurements of solar neutrinos. The results presented here will be the technical foundation for the solar neutrino physics program in DUNE and will inform future analyses and detector design optimization.

1.1 The Liquid Argon Time Projection Chamber

The Liquid Argon Time Projection Chamber (LArTPC) technology is based on the use of liquid argon as both the target and detection medium for neutrino interactions. It was proposed by Carlo Rubbia in 1977 [3] becoming a leading technology for neutrino detection due to its excellent spatial resolution, calorimetric capabilities, and scalability. In a LArTPC, when a neutrino interacts with an argon nucleus, it produces charged particles that ionize the argon atoms along their path. The liberated electrons drift under the influence of an electric field towards a readout plane, where they are collected and recorded as signals. The readout plane is typically segmented into multiple layers (induction and collection planes) that allow for a 2D reconstruction of the event. By combining the information from these planes with timing information, a full 3D reconstruction of the neutrino interaction can be achieved. Fulfilling the fundamental principle of a Time Projection Chamber (TPC). Additionally, liquid argon produces scintillation light when ionized, which can be detected by a Photon Detection System (PDS).

1.2 DUNE Far Detector

The DUNE FD will consist of four independent modules, with the first two (Phase I) based on the single-phase LArTPC technology, which is the most mature and well-understood design for large-scale neutrino detectors. The remaining two modules (Phase II) are planned to include one enhanced version of the Phase I modules and an additional *module of opportunity* that could be based on a different technology and/or design to optimize sensitivity to specific physics topics.

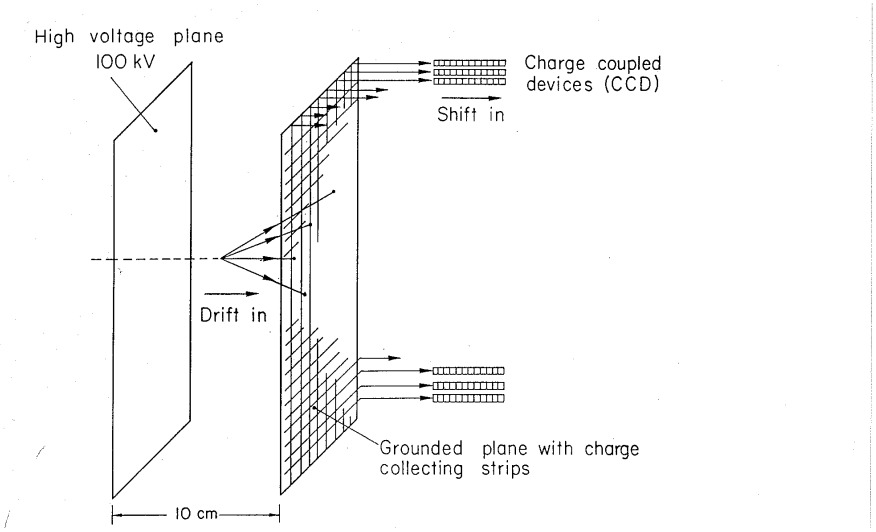


Figure 1: Schematic of the original liquid argon time projection chamber (LArTPC) showing the readout plane [3].

The FD modules will be located at the Sanford Underground Research Facility (SURF), which has 4700 m.w.e. of overburden, significantly reducing the rate of cosmic backgrounds and allowing for searches for low-energy astroparticle measurements. The PIP-II accelerator upgrade at the Fermi National Accelerator Laboratory (Fermilab) will provide a new neutrino beam directed at SURF to study long-baseline neutrino oscillations. In the first Phase of DUNE, two distinct designs (or geometries) will be implemented: the HD and the VD designs (see fig. 2), named after the orientation of their drift fields. These designs will allow for a comprehensive study of neutrino interactions and oscillations, as well as searches for proton decay and supernova neutrinos, contributing to our understanding of fundamental physics and the universe.

The HD's active volume will extend 14.4 m along the drift coordinate (x), 12 m along the vertical coordinate y, and 58.2 m along the beam-aligned coordinate (z). This module will be separated into four distinct drift volumes alternating three Anode Plane Assemblys (APAs) and two Cathode Plane Assemblys (CPAs), which are set to a drift potential of 180 kV, producing an electric field of 500 V cm^{-1}. The HD's APAs are equipped with three sets of sensing wires, all with a 5 mm pitch and with induction wires oriented at ± 37.5 degrees. The induction wires are positioned closest to the open drift volume, observing the signal as the free electrons transition through the layers. The rear *collection* wires absorb the charge of the drift signal and prevent the distortion of the electric field. This results in three semi-independent *views* of a neutrino interaction that can be combined to create time-dependet 2D representations of charged particle interactions in the detector. These two sets of induction wires wrap around the APA's centrally positioned collection wires, allowing the simultaneous observation of two drift volumes. This reduces the overall channel count but requires additional software computation to disambiguate the signal's interaction drift volume.

The FD VD module simplifies the more traditional HD design by dividing the active volume into just two drift volumes of dimensions 13.5 m $\times$ 6.5 m $\times$ 60 m. This has been made possible by advances in the Liquid Argon (LAr) purification system that allow for longer drift distances without significant charge loss. The module is divided horizontally by the cathode, denoted simply as *Cathode Plane*. The equivalent of the CPA in the VD module is called the Charge Readout Plane (CRP) and, differently to the thin and free frame-tensioned wires of a traditional LArTPC, its building block is a perforated Printed Circuit Board (PCB) with flat readout strips with a 3.125 mm pitch. Similarly to the case of the free wires in the HD design, these

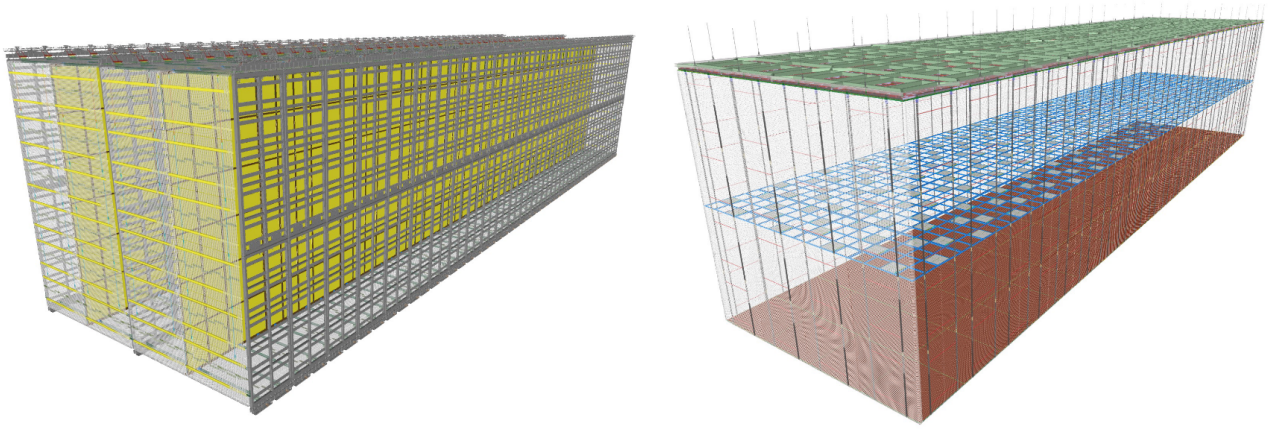


Figure 2: (Left) DUNE FD HD module. (Right) DUNE FD VD module.

perforations allow for the drifting electrons to pass by two induction PCB-planes, before being collected. This makes the analysis and simulation results very compatible between these two modules. The coordinate system of the VD geometry has been rotated by 90° along the z-axis to maintain the x coordinate along the drift direction. The induction strips are oriented at ±30 degrees with respect to the beamline, providing three views of the interaction. The VD module has a longer drift distance of 7 m and a lower electric field of 450 V cm^{-1}, which requires a higher cathode voltage of 350 kV. This design is more cost-effective and easier to construct, but, as we will demonstrate in the following chapters, it has worse performance in terms of drift coordinate reconstruction, energy resolution and therefore background rejection compared to the HD design.

DUNE FD Comparison

Parameter	HD Geometry	VD Geometry
LAr Mass	17.5 kt	
Inner Cryostat Volume	15.1 m × 14 m × 62 m	
Active LAr Mass	13.5 kt	14 kt
Nominal LAr Mass	10 kt	12.1 kt
Anode Modules	150 APAs	80 CRPs
Cathode Modules	300 CPAs	80 Units
Drift Length	3.6 m	7 m
Drift Field	500 V cm^{-1}	450 V cm^{-1}
TPC Voltage	−180 kV	−350 kV
TPC Views	3-view (± 37.5°)	3-view (± 30°)

Table 1: Key parameters of the HD and VD FD geometries.

Table 1 summarizes the key parameters of the HD and VD geometries. Given the nature of the LArTPC technology, in both geometries, the y and z coordinate reconstruction will follow from the TPC readout, together with a time-stamp. For beam events, the remaining x coordinate reconstruction follows from a simple calculation of the beam trigger. Note that for non-beam events (such as astrophysical neutrinos), the interaction vertex along the drift volume cannot be reliably reconstructed. To expand the reconstruction capabilities of the experiment, both detectors incorporate a PDS which makes use of X-ARAPUCA (XA) modules [4] distributed over the APA (cathode and membrane planes) for HD (VD). These provide an additional and instantaneous light signal for any interaction which, provided a correct match from the TPC readout is found, can be used to reconstruct the drift coordinate of low-energy interacting particles. This

enables background fiducialisation along the drift direction.

2 DUNE's Simulation Framework

The neutrino event simulation in DUNE follows the standard low-energy workflows of the Liquid Argon Software Framework (LArSoft) Monte Carlo (MC) framework. An example of the energy deposition and charge readout of a low-energy CC neutrino interaction in LAr can be seen in fig. 3.

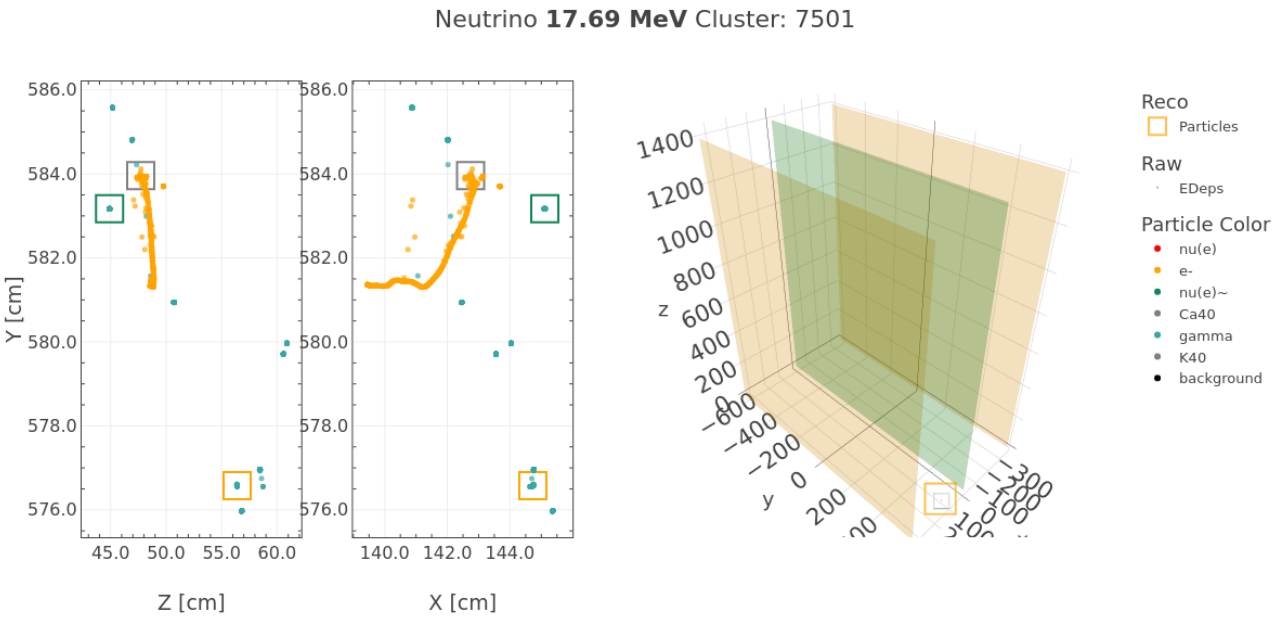


Figure 3: Example of a simulated ∼18 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron energy depositions (orange) and the associated de-excitation gammas (turquoise). The energy deposits from the truth information of the signal propagation are marked with dots and the reconstructed clusters with empty squares. (right) The reconstructed objects are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

The simulation workflow consists of five main stages. Events are initialized using the output of the MARLEY and Decay0 particle generators, propagated through the detector using the Geometry and Tracking v.4 (Geant4) software, which are implemented within the LArSoft framework. This framework provides the necessary tools for reconstructing the events using standard and custom algorithms, as well as, analysis tools to output the data in a standardized format. Finally, the data is further processed with custom tools for physics extraction and results.

2.1 Workspace Geometries

Due to the high computational load of low-energy productions (including backgrounds), the full scale geometries of the FDs are represented in simulation with smaller workspace geometries; designed to provide all possible edge and propagation effects of real neutrino and background interactions but reducing the detector's volume to contain all the signal of a single neutrino

event. In this work, we used two geometries: the HD workspace geometry (hd_1x2x6) and the VD workspace geometry (vd_1x8x14_3view_30deg). In both cases, the x and y coordinates are set to the same lengths as for the full drift volumes but the z coordinate is shortened roughly 1/4 and 1/3 of the full length, respectively. The HD workspace features a single central APA and two adjacent CPAs creating two drift volumes that are designed to emulate the central region of the HD detector. On the other hand, the VD workspace geometry consists of a single drift volume that represents the upper half of the detector (see fig. 4).

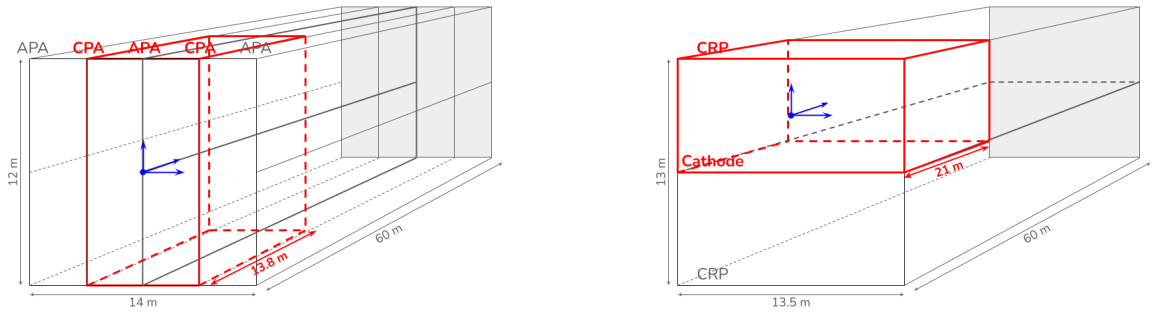


Figure 4: (left) The HD workspace geometry (hd_1x2x6_centralAPA) with a single central APA and a drift length of 3.6 m. (right) The VD workspace geometry (vd_1x8x14_3view_30deg).

The HD geometry is used in low-energy neutrino studies simulating two background scenarios: one without external backgrounds from the lateral faces of the cavern APA (hd_1x2x6_central APA) and another one with complete external background (hd_1x2x6_lateralAPA). Due to the arrangement of the TPC planes in the workspace geometry, the latter background scenario is required to implement the lateral cavern flux in the center of the detector (where the APA is placed). This creates a scenario where the central region of the workspace actually represents the most external faces of the detector's charge readout. In both scenarios, we simulate the full set of radiological backgrounds, together with top, bottom, head, and tail external backgrounds. In the hd_1x2x6_lateralAPA scenario, the lateral cavern flux is also included in the central APA region.

Workspace Comparison

Parameter	HD Workspace	VD Workspace
#Drift Volumes	2	1
Active Volume	7.2 m × 12 m × 14 m	6.5 m × 13.5 m × 21 m
Active Mass	1.6 kt	2.5 kt
Anode Modules	12 APAs	28 CRPs
Cathode Modules	24 CPAs	28 Units

Table 2: Key parameters of the HD and VD workspace geometries used in the simulation.

For the VD geometry, we consider two scenarios: the nominal scenario, which represents the upper half of the VD module (vd_1x8x14_3view_30deg_nominal), and a shielded scenario that assumes additional reduction of backgrounds in the form of altered fluxes (vd_1x8x14_3view_30 deg_shielded). The latter scenario is likely to represent the lower half of the VD module, where shielding is more suitable from and engineering point of view. This does not require any changes in the geometry itself, but only in the background fluxes used in the simulation. In both cases, we simulate the full set of radiological backgrounds, together with top, head, tail, and lateral external backgrounds (bottom external backgrounds are not simulated in the VD geometry due to fact that we are only simulating half of the detector's volume).

2.2 Event Generation

The simulation of low-energy neutrino interactions in DUNE relies on a comprehensive workflow that encompasses several stages, from the event generation to the final analysis data. The workflow is designed to produce realistic simulations of neutrino interactions, accounting for the complex physics processes involved and the detector response. The main stages of the simulation workflow are illustrated in fig. 5 and include:

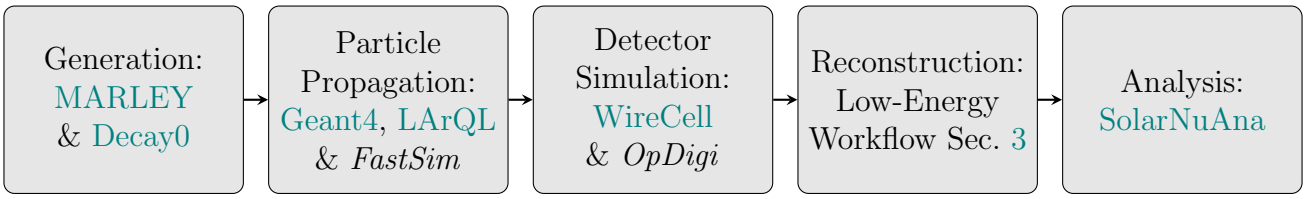


Figure 5: Simulation workflow stages for neutrino event reconstruction in DUNE. The pipeline progresses from event generation through particle propagation, detector simulation, and reconstruction to final analysis.

2.2.1 Neutrino Interactions

The first step in simulating neutrino interactions relies on a comprehensive nuclear model of the possible processes. For low-energy events, this is the MARLEY, an event generator developed to simulate the nuclear interactions of low-energy neutrinos in the range of a few to tens of MeV. It is particularly suited for simulating interactions relevant to solar and supernova neutrinos in LAr detectors. In this work we focus on the CC process (although Elastic Scattering (ES) and Neutral Current (NC) have also been implemented):

$$\nu_e + {}^{40}\mathrm{Ar} \rightarrow e^- + {}^{40}\mathrm{K}^* \,. \tag{1}$$

Upon an electron neutrino capture on an argon nucleus and subsequent electron release, the excitation of the residual ${}^{40}\mathrm{K}^*$ leads to the emission of de-excitation γ-rays, which can produce observable signals (blips) in the vicinity ($<$1 m) of the main electron track (see fig. 3). MARLEY incorporates a gamma-decay schemes to describe these processes and feeds the resulting daughter particles into LArSoft, which is used to produce a MC input for physics studies. It simulates primary interaction kinematics, the resulting nuclear state and the gamma-cascades from nuclear de-excitation, as well as, other secondary processes (such as neutron or proton emission). In DUNE's physics analyses, the most important signature from these interactions is the already mentioned primary electron track. The resulting energy of these is strongly dependent on the de-excitation scheme of the residual nuclear state (see fig. 6).

The average energy transfer and binding energy of the CC interaction are displayed. One can observe how the gamma channel starts dominating the interaction up to $\sim$10 MeV, and above that threshold, the main electron takes over the fractional energy. This trend is accompanied by a continuous increase in average kinematic threshold caused by the production of additional secondary particles. The most concerning secondary production, from the experimental point of view, is the emission of neutrons; which happens above $\sim$15 MeV, and leads to a decrease in visible energy that is unlikely to be recovered under realistic reconstruction scenarios. All of these effects combine to a strong energy dependence of the deposited and visible energies, which should be taken into account in our signal calibration. The same can be said about the

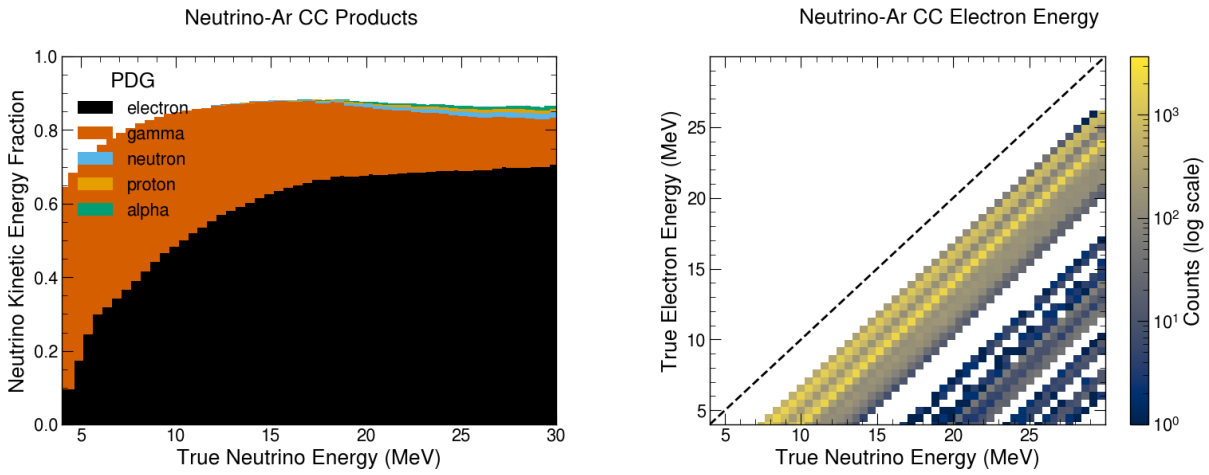


Figure 6: MARLEY's computation of CC neutrino interaction. (left) Daughter fraction of the original kinetic energy. (Right) Neutrino-electron energy correlation in a CC interaction as predicted by MARLEY.

missing energy due to nuclear binding energy losses, which are also energy dependent and lead to a non-negligible bias in the reconstructed neutrino energy if not properly accounted for. This is particularly important at higher energies, where the binding energy losses can represent a significant fraction of the total neutrino energy.

It is important to mention at this point that a typical low-energy MC production includes a full propagation of the backgrounds described in section 2.2.2. These additional particles are produced in their specific rates and concentrations according to computations out of the scope of this publication. All particle kinematics and detector placement happen in the so-called *gen* stage of the simulation. All produced background particles follow the same workflow chain as the signal particles and contribute to the ultimate smearing and resolution of the reconstructed objects. In order to account for all possible matching issues one full drift-time needs to be simulated before the neutrino makes its interaction (at $t = 0$). After this, the simulation last for another full drift-time to allow neutrinos from all regions of the detector to drift and be collected onto the corresponding readout plane. For the purpose of this publication, neutrinos are produced with isotropic positions and directions across the full active volume; and with a flat energy spectrum from 4 MeV up to 30 MeV.

2.2.2 Background Model

One of the most studied backgrounds for LAr detectors is Radon (^{222}Rn), which is a noble gas produced in the decay chain of ^{238}U [5, 6, 7]. As a gas, it can diffuse into the active volume of LAr detectors, where its decay daughters accumulate on surfaces or remain suspended, emitting beta, gamma and alpha radiation (see Appendix section 7). Alpha particles have a very short range in LAr, but due to their high energy and quenching, they produce significant light signals in the detector, which can be misidentified as low-energy neutrino interaction candidates.

The measurement of low-energy neutrinos in DUNE, particularly those from the Solar ^{8}B chain (<19 MeV), requires a thorough understanding of intrinsic (emitted from the LAr), component-induced (emanating from detector components) and external (penetrating from the cavern) backgrounds. Understanding the contributions from these particles and decaying isotopes is crucial in achieving a correct vertex and energy reconstruction required for low-energy neutrino

analyses. Apart from the previously mentioned radon chain, a comprehensive list of background sources has been implemented in DUNE's simulation. Presenting DUNE's full background model is out of the scope of this work. Hence, we focus on providing a brief summary while discussing the features that have the biggest impact on the low-energy neutrino reconstruction analyses.

- **External Backgrounds**: External radiation originates outside the detector's cryostat. The flux is divided into two kinds of particles that manage to penetrate into the detector:
  - **Cavern Neutrons**: Produced by (α, n) reactions in surrounding rock or via cosmic-ray spallation, neutrons are emitted up to $\sim$10 MeV in kinetic energy. They can enter the active volume where they undergo ES until they capture producing a gamma-cascade which can mimic the topology of a CC neutrino interaction. Some neutrons can be captured on the cryostat and other detector components contributing to the environmental particle flux with high energy gammas.
  - **Environmental gammas**: High-energy photons from the rock or shotcrete surrounding the detector, typically originating from natural radioactivity. The highest rate contributor to this flux in DUNE's caver at SURF is the shotcrete covering the walls. Low-energy gammas (<2.7 MeV) entering the active volume have a huge flux contributing mostly to worsen the signal reconstruction by mimicking the de-excitation gammas of the CC interaction or making the topological distinction between ES and CC less reliable. The higher energy external gammas (2.7 < E <14 MeV), are produced by γ-decay of isotopes like ^{208}Tl ($\sim$3.5 MeV) or ^{214}Bi ($\sim$3.27 MeV) and secondary nuclear reactions of (n, γ) interactions in certain geological environments (>6 MeV). These high-energy gamma rays are highly site-specific, depending on the local neutron flux and rock composition. Therefore, in situ measurements are fundamental to assess their impact and model accurately their contribution as fake signals to the different physics analyses.
- **Component-Induced Backgrounds**: These can arise from the materials used to construct the detector itself. Material screening and selection, as well as radon mitigation systems, are essential to reduce these sources. These include decay chains of long-lived isotopes introduced during fabrication or operation. Some of the most prominent are:
  - **Uranium (^{238}U)**: Produces alpha, beta, and gamma emissions, as well as daughters like ^{214}Bi and ^{210}Pb.
  - **Thorium (^{232}Th)**: Similar to the uranium chain, with prominent gamma emitters like ^{208}Tl (2.6 MeV).
  - **Potassium (^{40}K)**: Often present in plastics and glass (like the ones used for traditional photon detectors). It emits a 1.46 MeV gamma line.
  - **Radon (^{222}Rn)**: Present in atmospheric air and therefore very difficult to prevent from accumulation on detector and component surfaces during construction.
- **Intrinsic Liquid Argon Backgrounds**: When talking about the properties of DUNE's LAr we are generalizing the properties of (^{40}Ar) but certain radioactive isotopes are either present naturally in atmospheric argon or may be introduced as impurities:
  - **Argon (^{39}Ar)**: A beta emitter with an endpoint energy of 565 keV and a half-life of 269 years. It dominates the low-energy spectrum in atmospheric argon with an activity of $\sim$1 Bq kg^{-1}. Together with the external gammas, this background

contributes to the smearing of signals and worsens the hit reconstruction efficiency of the TPC by significantly altering the waveforms and providing fake signal candidates for plane-matching.

- **Argon (^{42}Ar) and Potassium (^{42}K)**: Rare isotopes that decay via beta emission. ^{42}K emits an electron ($Q \sim 3.5$MeV), which is particularly problematic due to its similarity to low-energy solar neutrino events.
- **Krypton (^{85}Kr)**: A beta emitter with a Q-value of 687 keV, introduced via contamination from air or during purification. Often requires cryogenic distillation to remove. It contributes to the low-energy background spectrum, similar to ^{39}Ar.
- **Radon (^{222}Rn)**: In addition to emanation from components, radon can dissolve in liquid argon and contribute to intrinsic backgrounds through its decay chain that can appear throughout the full LAr volume.

In DUNE, LAr purification and recirculation installations are planned to ensure constant operating conditions. A detailed background model informed by in situ measurements, simulations, and radiopurity assays has been developed and is under constant revision to ensure accurate predictions of the background fluxes in the LAr active volume. The model is implemented in the LArSoft [8], the software framework used for DUNE's simulations, and is used to generate realistic event samples for detector response studies and physics analyses.

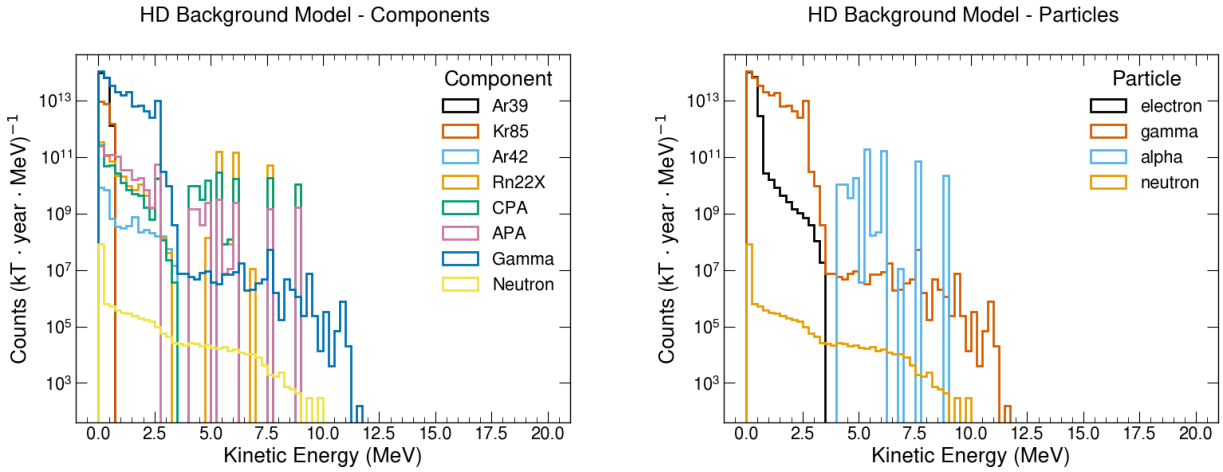


Figure 7: Simulated particle flux from DUNE's Background Model version 3. (Left) Divided by the background component. (Right) Divided by particle type. The external gamma and neutron flux are computed for the inner surfaces of the cryostat, where they are further propagated and combined with the rest of the intrinsic backgrounds.

Figure 7, shows the latest simulated particle flux from the background model (v. 3.6) on top of the predicted solar spectrum as a function of the particles' kinetic energy. The left panel shows the flux divided by background components, while the right panel shows it divided by particle type. The flux is dominated by external gammas, argon and krypton at low- and external gammas and neutrons at high-energies. These figures highlight the importance of understanding these backgrounds by highlighting the huge difference in flux between the low-energy neutrino signal and the background components. Without a proper energy calibration, the smearing of the background components can lead to a significant loss of signal efficiency and a degradation of the energy resolution.

2.3 Particle Propagation

Following the generation of the primary neutrino interaction via the MARLEY generator for low-energy solar neutrino events, the resulting final-state particles are propagated through the liquid argon medium using a Geant4 simulation [9]. This so called *g4* stage, models the particle-matter interactions, including energy loss, scattering, bremsstrahlung, photon production, hadronic interactions and more. This happens according to a physics list tailored for these low-energy processes (larg4services). The result of this first step is a detailed list of daughter particles and their trajectories, along with the energy deposited in the liquid argon at each step of the particle's path.

As these traverse the liquid argon, charged particles will lose energy primarily through ionisation, producing free ionisation electrons and excitation of argon atoms, which subsequently decay and emit prompt scintillation photons in the Vacuum Ultraviolet (VUV) range (128 nm). These processes are central to the signal generation in a LArTPC. For computational efficiency, the simulation does not explicitly track every ionisation electron or VUV photon. Instead, energy loss along particle trajectories is recorded as a set of energy depositions, or `edeps`, which include position, time, energy, and associated particle identity. Similarly, the number of scintillation photons produced in each energy deposition step is recorded as a secondary output, commonly referred to as `simphotons`. These `edeps` and `simphotons` are then effectively propagated through the LAr volume with custom analytical algorithms.

The computation of light and charge production for the DUNE FD utilizes the LArQL (Liquid Argon Charge and Light) model for correlated signal generation. LArQL is a phenomenological approach implemented to provide an improved description of the anti-correlation between free ionization charge (Q) and scintillation light (L). It is particularly effective for DUNE simulations as it covers electric field ranges from $0\,\mathrm{kV\,cm^{-1}}$ to $0.75\,\mathrm{kV\,cm^{-1}}$ and energy densities (dE/dx) from $2\,\mathrm{MeV\,cm^{-1}}$ to $40\,\mathrm{MeV\,cm^{-1}}$. The model is built upon the unitary charge-light equation of the traditional Birks' model:

$$Q\left(\frac{dE}{dx},\mathcal{E}\right) + L\left(\frac{dE}{dx},\mathcal{E}\right) = N_i + N_{ex} \tag{2}$$

where N_i is the number of ionization electrons initially generated ($N_i = 1/W_{ion}$) and N_{ex} is the number of primary excitons.

Considered the evidence of light reduction at $\mathcal{E} \to 0$ for lower $\frac{dE}{dx}$ values [10], one can establish the fraction of missing photons $(1-\eta_0)$ and from this the fraction of escape electrons (χ_0):

$$\eta_0 = \frac{1-\chi_0+N_{ex}/N_i}{1+N_{ex}/N_i} \quad \Rightarrow \quad \chi_0 = (1+N_{ex}/N_i)\cdot(1-\eta_0) \tag{3}$$

With this, the LArQL modification to the traditional Birks' model adds a term for these escaping electrons (χ_0):

$$Q_{\mathrm{LArQL}} = \frac{A_B}{W_{\mathrm{ion}}\cdot\left(1+\frac{k_B}{\mathcal{E}\rho_{\mathrm{LAr}}}\cdot\frac{dE}{dx}\right)} + \chi_0\left(\frac{dE}{dx}\right)\cdot f_{corr}\left(\frac{dE}{dx},\mathcal{E}\right)\cdot Q_\infty \tag{4}$$

where A_B and k_B are original Birks parameters, ρ_{LAr} is the LAr density and f_{corr} is an exponentially decreasing function that reduces the contribution of escape electrons as the field increases which is defined as:

$$f_{corr} = e^{-\mathcal{E}\cdot\left(\alpha \ln(\frac{dE}{dx})+\beta\right)^{-1}}. \tag{5}$$

Here, the parameters α and β are optimized through global fits of experimental data from ICARUS, MicroBooNE, and ARIS.

After the generation of the ionization charge and scintillation light in the *g4* stage, the simulation proceeds to propagate the photons wit a so called *FastSim* approach. The method relies on a *Photon Library* which is pre-computed using a standalone Geant4 simulation that deposits a fixed number of photons in each voxel of the detector volume and records the response (number of photoelectrons) for every photon detector. During the standard simulation, the software looks up the pre-calculated response for the voxel where light was produced, vastly reducing CPU time while still providing an accurate representation of the light collection efficiency across the detector volume [11].

2.4 Detector Simulation

In this, so called, *detsim* stage; after the effective production of ionisation electrons and propagation of scintillation photons, the signal formation is modelled. This stage takes as input the energy depositions (`edeps`) and the associated scintillation photons (`simphotons`) from the particle propagation step.

2.4.1 TPC Signal

WireCell is a vast software toolkit that provides a comprehensive simulation of the TPC signal formation and reconstruction [12]. It is defined as a tomographic automated 3D neutrino event reconstruction method for LArTPCs. Unlike paradigms that perform pattern recognition on 2D views first, WireCell reconstructs the ionization density in 3D space prior to particle identification.

In DUNE's current simulation processes, the `edeps` are passed as a distributions of charge (q_i) at specific space-time points (r_i, t_i) to produce a realistic detector responce. Conceptually, the simulation operates through five key functional stages that are intertwined in a convolutional chain [13, 14]:

1. **Deposition (`edeps`)**: Represents the initial distribution of ionization electrons in space and time.
2. **Drift**: Models the transport of electrons to the anode, accounting for **attenuation** (due to impurities) and both **longitudinal and transverse diffusion**.
3. **Duct (Field Response)**: This is a convolution of the position-dependent **field response function** (F) and the electronics response. It utilizes Garfield-simulated field responses based on Ramo's theorem to model current induction, including **long-range effects** on neighboring wires.

4. **Digitization**: Models the front-end electronics based on specific sampling rates (e.g., 2 MHz), resolution, and dynamic range to produce ADC waveforms.

5. **Noise**: Inherent electronics noise is added using a data-driven analytical model, often employing a Rayleigh-distributed amplitude spectrum.

The WireCell simulation has been rigorously validated against data from experiments such as MicroBooNE and ProtoDUNE-SP. By modeling signal formation from first principles, it provides a crucial foundation for developing 3D reconstruction algorithms and evaluating systematic uncertainties in experiments like DUNE.

To improve charge extraction, the waveforms undergo a *deconvolution* procedure, where the electronics response and noise spectrum are inverted in frequency space to recover a cleaner signal representation. The deconvolved waveforms are evaluated under a Region Of Interest (ROI) finder algorithm and trimmed for subsequent evaluation. This last step reduces the data volume by focusing only on regions with significant signal activity and trimming all information around the signal peak that is compatible with baseline noise.

2.4.2 PDS Signal

Simultaneously, the scintillation photons are detected by the PDS, composed of XAs. The `simphotons` from Geant4 are converted into raw signals. Similarly to the TPC reconstruction, the PDS response is also emulated with realistic timing resolution, dark noise, and electronic gain amplification. Simulated waveforms are produced for each optical channel and overlaid with noise. Also in this case, fixed ROI windows are defined after the waveforms surpass a certain threshold, indicating potential scintillation events. In contract to the TPC ROIs, the waveforms of the PDS can be stored and process entirely due to the limited number of channels so entire 16 µs long ROIs are stored.

Once the number of arriving photons as a function of time is obtained per Photon Detector (PD) channel, the digitization of the signal is performed. The generation of optical waveforms takes into account the expected Photon Detection Efficiency (PDE) of the XAs, the single Photoelectron (PE) signal response, Silicon Photomultiplier (SiPM) crosstalk, dark count rate, and electronic noise. Those waveforms go through a peak finding algorithm. For the VD case, this is simply done by looking for peaks above a threshold (usually corresponding to 1.5 PEs) and assigning the number of PEs as the integral of the peak divided by the area corresponding to a single PE. However, the HD case presents additional complexity, as the single PE shape shows a sizeable undershoot which requires a deconvolution to be performed before obtaining the number of PEs in a given peak.

3 Low-Energy Event Reconstruction

The signal waveforms are passed forward to the reconstruction routines. A peak finding algorithm fits Gaussian distributions to the signal waveforms, where each successful fit is formed into an object known as a TPC *hit*. The area under all the TPC hits is proportional to the energy deposited by the charge particles as they propagate through the liquid argon volume. For the horizontal drift processing chain, a disambiguation routine is then applied to the induction

plane hits to deduce which drift volume they originated from. This routine combines spatial and timing information of all hits in all planes simultaneously to maximize performance. The TPC hits are then passed forward to the higher-level reconstruction routines.

As a next step, after the generation, propagation and detector simulation of the MC workflow, the reconstruction (*Reco*) step for these studies includes basic hit-finding algorithm and clustering schemes that applies both to the TPC and PDS. This *Reco* stage typically is subdivided into two:

Reco1: Conceptually, this part of the process is designed to provide the common building blocks for subsequent reconstruction stages: the `hits`. These are localized charge or photon depositions in the TPC or PDS, respectively. As quanta of information, the algorithms that produce them are optimized for efficiency and accuracy by a full team of LArSoft developers. The `hits` used in this work are the standard LArSoft objects produced from the main reconstruction workflow and common to all DUNE studies. Nevertheless, higher-level clustering algorithms have been specifically developed to group these `hits` into more complex objects that can be used for low-energy event reconstruction.

Reco2: This second part of the reconstruction process focuses on higher-level object formation and the association between TPC and PDS systems.

The clustering algorithms developed for this purpose are detailed in sections 3.1 and 3.2 and can be found in the following repositories: duneopdet & dunereco; as well as the associated producer modules: SolarOpFlash_module.cc & SolarCluster_module.cc. A custom solar workflow leading to the production of these reconstructed objects has been included in the standard production workflow files: solar_reco_dune10kt_1x2x6.fcl & solar_reco_dunevd10kt_1x8x14_3view_30deg.fcl.

3.1 TPC Clustering

The clustering of TPC hits is performed using a simple algorithm based on the successive scanning of a time-ordered list of hits. The algorithm starts by taking the first hit in the list and grouping it with all hits that are within a certain distance in both channel and time. This process is repeated for each hit until all hits have been processed. The resulting clusters are then filtered based on their total charge, and only those above a certain threshold are kept for further analysis.

The clustering scheme is illustrated in fig. 9. The choice of the distance threshold is crucial, as it determines the size and shape of the clusters. A smaller threshold will result in more, smaller clusters, while a larger threshold will produce fewer, larger clusters. The optimal threshold is determined based on the expected signal characteristics and the background conditions of the detector. In this work, we found that a threshold of 3 adjacent channels and a time window of 12.5 µs (25 tick) provides a good balance between cluster completeness and purity in the context of background. The resulting clusters contain the weighted average time and wire-position, as well as the total charge, which are used for subsequent reconstruction stages.

Figure 9 shows an example of the clustering process for a simulated neutrino interaction in the HD geometry. The hits are represented as black squares, and the dashed circles indicate the adjacency region for each hit. The colors represent the hits that have been grouped together into clusters. The resulting clusters contain the weighted average time and wire-position, as well as the total charge, which are used for subsequent reconstruction stages.

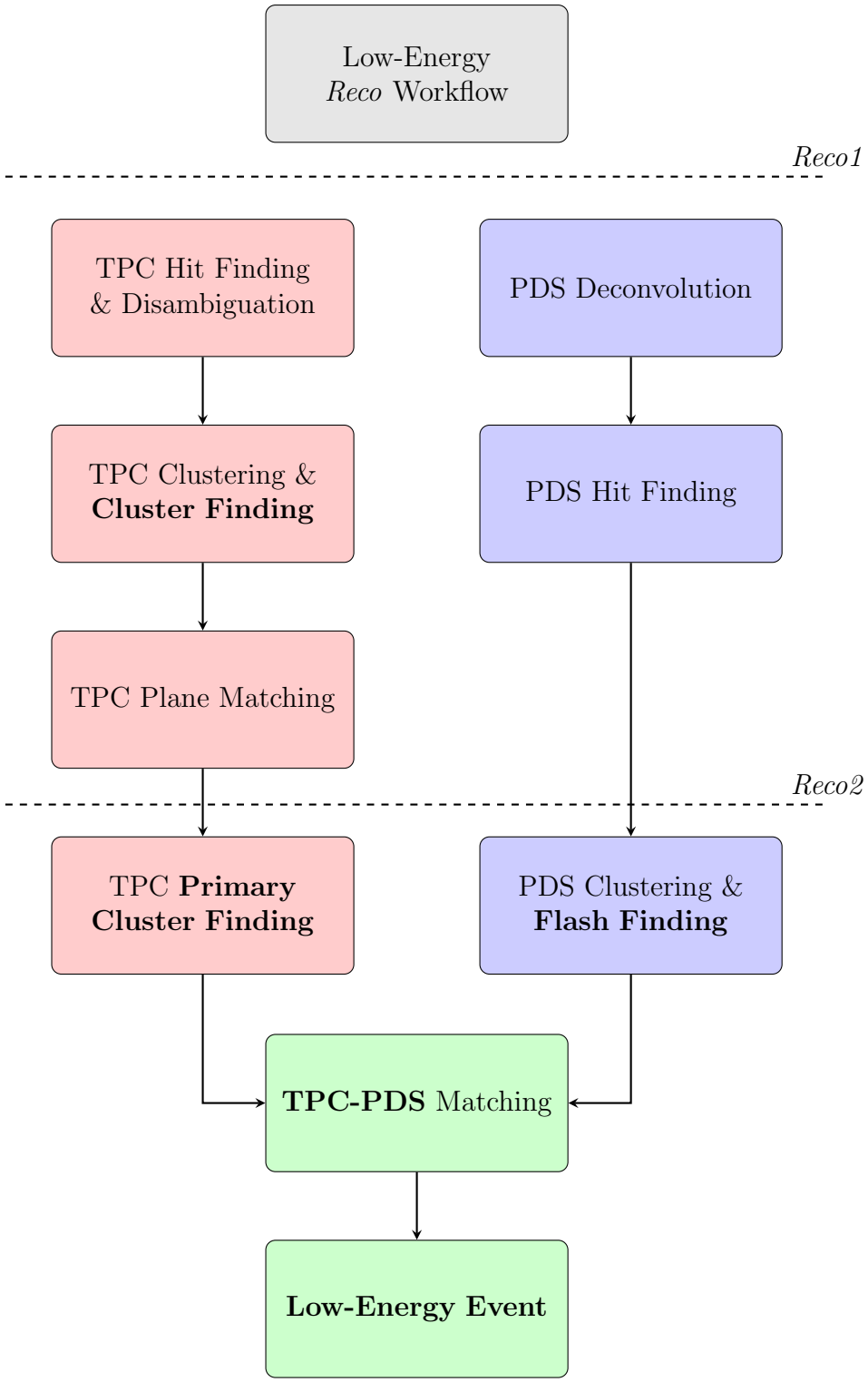


Figure 8: Schematic of the low-energy reconstruction workflow, divided into two main stages: *Reco1*, which focuses on hit finding and clustering for both TPC and PDS systems; and *Reco2*, which emphasizes higher-level object formation and the association between TPC and PDS systems to ultimately form low-energy event candidates.

Clusters are associated to others based on their relative charges and radial distance. The adjacent cluster finding scheme is illustrated in fig. 10. For the radial distance comparison between clusters, the difference in their arrival times is converted to the missing spacial coordinate by taking into account the known drift velocity in LAr. Notice that this computation only holds true for correlated energy deposits (e.g. nuclear de-excitation gammas from a CC neutrino

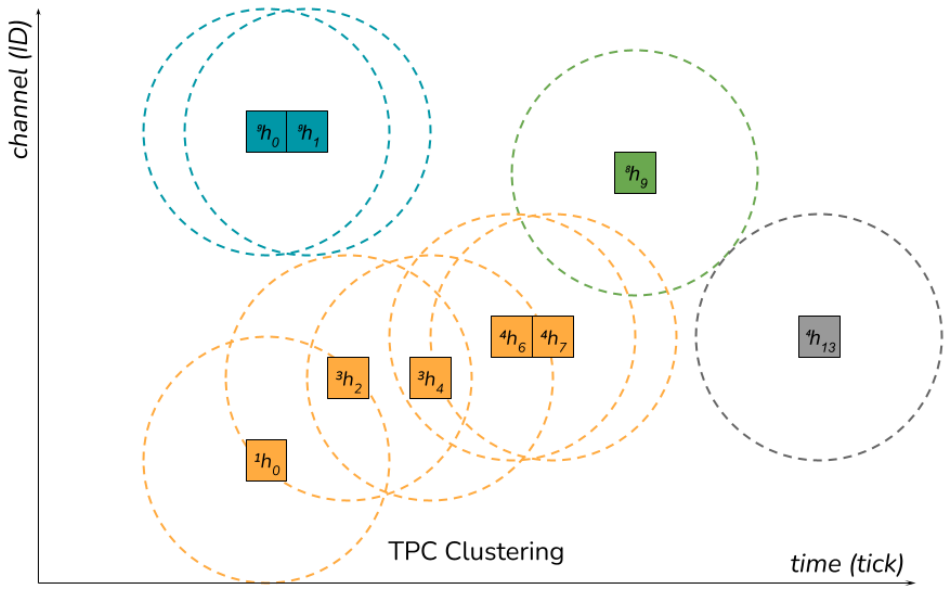


Figure 9: TPC Clustering scheme based on the successive scanning of a time-ordered `hits` (represented as black squares). The dashed circles represent the "adjacency" region for each hit. Colors are used to represent `hits` that have been grouped together. For simplicity, unitary distance in channel (ID) and time (ticks) are shown to be equivalent.

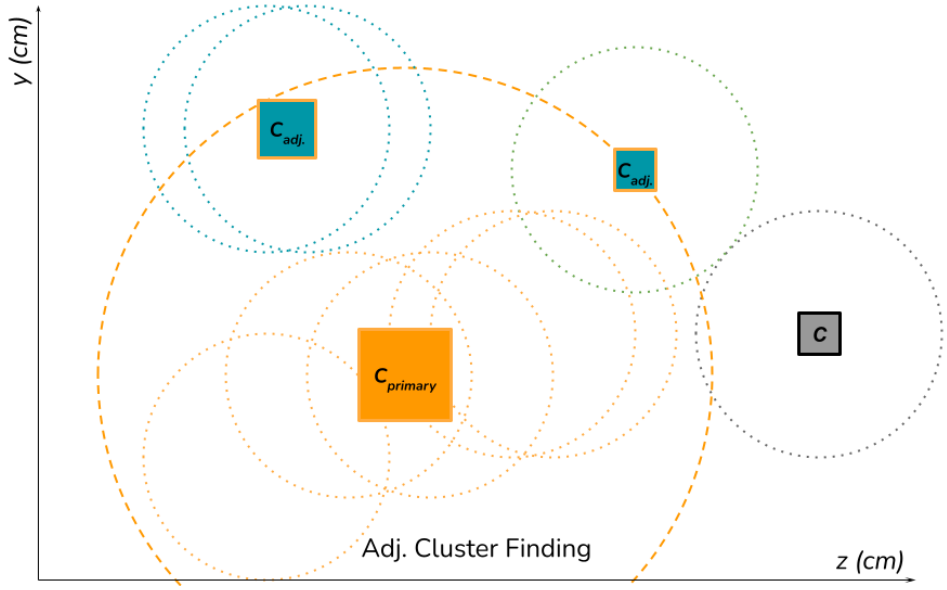


Figure 10: Adjacent cluster finding scheme. The size of the clusters represents their total charge. The dashed circle represents the 1 m radial distance used to find adjacent clusters around the primary cluster. Once found, these clusters are grouped together with the primary cluster to form a low-energy event. Clusters cannot be considered adjacent to more than one primary cluster.

interaction in LAr following the emission of a main electron). In the case of non-correlated background events occurring throughout the full detector volume, their distance along the drift coordinate with respect to other clusters cannot be truthfully determined. From a reconstruction point of view, this leads to the presence of a constant background floor in the vicinity of any signal cluster that corresponds to the projected activity in the interacting drift volume (see tables 6 and 7). These clusters are divided into:

- **Primary**. Clusters with charge bigger than any other's cluster charge up to 1 m radial distance. In physics studies, such as the solar neutrino analysis, these clusters can be both signal (from neutrino interactions) or background (neutrino event candidates). Here we will focus only on primary clusters from neutrino interactions, but the calibration and reconstruction techniques described here are equally applicable to background-induced primary clusters.
- **Adjacent**. Clusters with charge smaller than the one of a primary cluster closer than 1 m radial distance.

Depending on the particle producing the energy deposition, adjacent clusters can be further divided into three categories based on their origin. In the following we will distinguish between three types of adjacent clusters:

- **Signal**. These are clusters produced by secondary gammas emitted in the de-excitation of the final state nucleus after the neutrino interaction. These clusters are correlated in time and space with the primary cluster.
- **Intrinsic Background**. These are clusters produced by background sources that are intrinsic to the LAr or detector materials. These clusters are uncorrelated in time and space with respect to the primary cluster, but their rate is uniform throughout the detector volume and does not depend on the forementioned background scenarios for each of the geometries.
- **External Background**. These are clusters produced by background sources that originate outside the detector active volume (e.g. gammas from the cavern walls, neutrons from the surrounding rock, etc.). These clusters are equally uncorrelated in time and space with respect to the primary cluster and their rate depends on the background scenario for each of the geometries.

This distinction is important for the reconstruction of the neutrino energy, as signal adjacent clusters can be used to improve the energy resolution, while background adjacent clusters can significantly worsen it. The presence of background adjacent clusters is one of the main challenges for the reconstruction of low-energy neutrino interactions in LAr, and different strategies are studied in this work to mitigate their impact on the energy reconstruction.

3.2 PDS Clustering

Similarly to the TPC clustering, the PDS signals are also clustered in order to reconstruct the flashes. The flash finding scheme is illustrated in fig. 11. The algorithm takes into account the time and channel proximity of the `ophits` to cluster them into flashes. Each flash is then assigned a vertex and a time, which are used for the flash-matching with the TPC clusters. The resulting flash contains information about the total number of PEs, the weighted average time of the flash, and the flash vertex (computed as the weighted average position of all XA channels). The choice of the time and channel proximity thresholds is crucial, as it determines the size and purity of the flashes. A smaller threshold will result in more, smaller flashes, while a larger threshold will produce fewer, larger flashes. The optimal thresholds are determined based on the expected signal characteristics and the background conditions of the detector.

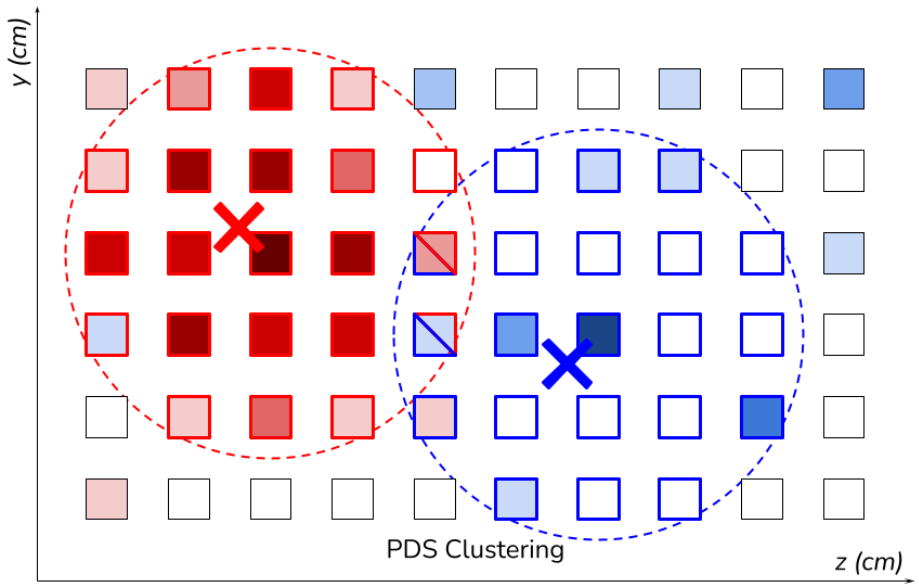


Figure 11: Flash finding scheme. Optical channels (XAs) are represented as black squares. The darkness of their filling represents the number of PEs detected per `hit`. The color represents the `ophit`'s precedence: signal (red) and background (blue). The dashed circle shows the `ophits` grouped into a single flash. Notice how the algorithm allows for the same `ophit` to be part of more than one flash.

In the case of the VD, the clustering is performed independently for each PDS plane. In this work, we found that a time window of 100 µs and a radial distance of 1 m provides a good balance between flash completeness and purity in the context of background. The parameters used for the flash finding are shown in table 3. Note that these parameters will be held the same for the different background configurations, while the matching parameters can be optimized for each configuration to maximize the matching efficiency.

Parameter	**HD**	**VD**
`OpFlashAlgoMinTime`	0.16 µs	1 µs
`OpFlashAlgoMaxTime`	0.32 µs	1.6 µs
`OpFlashAlgoRad`	600 cm	600 cm

Table 3: Flash-finding parameters for HD and VD geometries.

The flashes are produced from `ophits` by clustering these PDS signals in time and channel proximity. All `ophits` within a given time window (`OpFlashAlgoMinTime` - `OpFlashAlgoMaxTime`) and distance (`OpFlashAlgoRad`) are grouped together around a single trigger `ophit`, which is defined to be the one with the highest number of PEs. The clustering is performed independently for each PDS plane in the VD detector. The resulting flash contains information about the total number of PEs, the weighted average time of the flash, and the flash vertex(computed as the weighted average position of all XA channels).

Figure 12 shows the flash-finding efficiency for the HD and VD geometries as a function of the distance to the PDS plane and the true neutrino energy. The flash-finding efficiency is defined as the fraction of neutrino interactions that produce a flash according to the parameters specified in table 3. The efficiency is higher for the HD geometry due to its smaller drift volume, which results in a higher probability of detecting the scintillation light produced by the neutrino interactions. The efficiency also increases with neutrino energy, as higher energy interactions

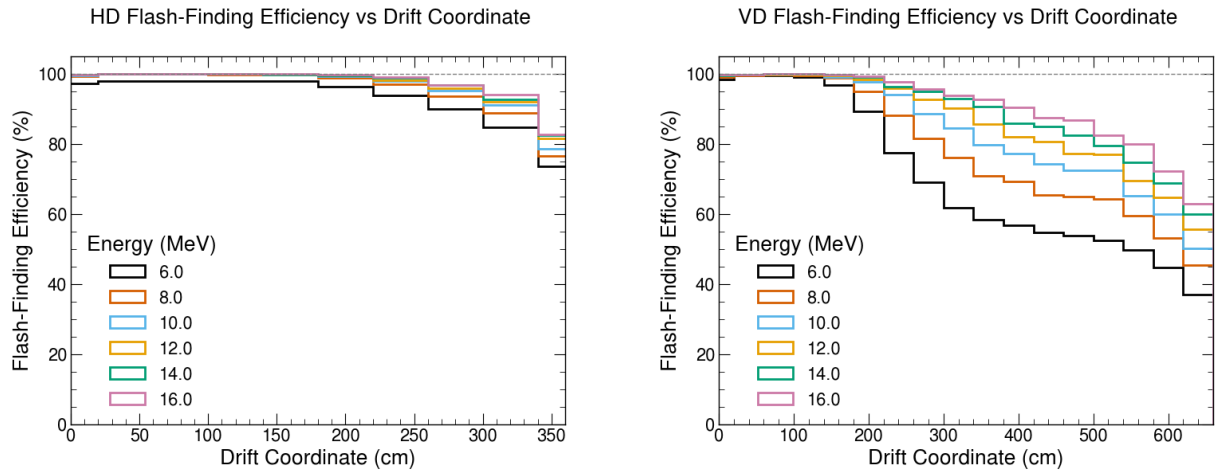


Figure 12: Flash-Finding efficiency for the HD (left) and VD (right) geometries. Results are shown as a function of distance to the PDS plane and true neutrino energy.

produce more scintillation light, which is easier to detect and cluster into flashes.

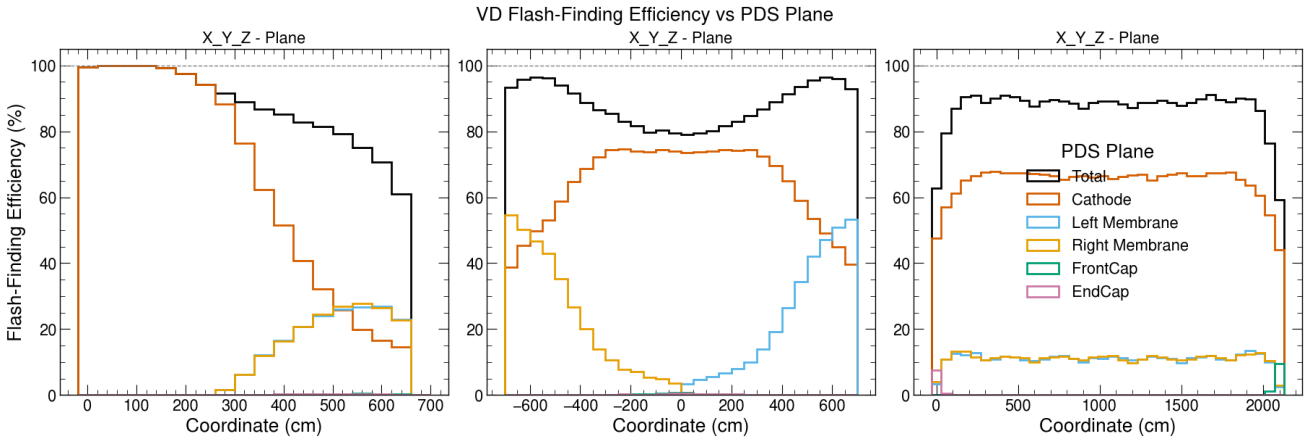


Figure 13: Flash-Finding efficiency for the VD geometry. Results are shown as a function of (left) distance to the PDS plane, (center) true neutrino energy and (right) PDS plane match.

Once the set of flashes is formed, the matching is performed by evaluating the distance between the TPC vertex and the PDS flash, as well as their time difference. Flashes can be formed from signal or background activity. For any given flash, there is a non-negligible chance clustering pileup (where `ophits` formed from background activity get clustered into flashes where the trigger `ophit` originates from signal) or channel pileup (where `ophits` themselves are formed from a mixture of simultaneously arriving signal and background photons) occur. Therefore, we distinguish between:

- **Signal flashes**: those that contain at least some `ophit` originating from the signal activity. In any realistic scenario, these flashes will be contaminated with at least some amount of photons originating from background activity. Signal flashes can be subdivided into:
  - **High Purity**: those that contain mostly `ophits` originating from the signal activity. Their reconstruction is likely to provide an accurate estimate of the true neutrino interaction time and good vertex reconstruction.
  - **Low Purity**: those that contain mostly `ophits` originating from background activity. Their reconstruction is likely to provide a smeared estimate of the true neutrino interaction time and poor vertex reconstruction.

- **Background flashes**: those that contain only `ophits` originating from background activity. These flashes can be mistakenly matched to a TPC cluster, leading to a wrong drift coordinate reconstruction. As these are not correlated in time with the neutrino interaction, they will lead to a random and flat drift coordinate assignment within the detector active volume.

This distinction will become important as we will see in section 5.2. Before attempting any full reconstruction based on these matching, it is important to understand the reach of the matching algorithm.

3.3 Flash-Matching

The final step in the event reconstruction process is the attempt to match the reconstructed primary TPC cluster with a corresponding PDS flash. Note that this step is not strictly required for the energy reconstruction, as the drift attenuation correction can be applied statistically. However, the matching is essential for the correct reconstruction of the signal's 3D position and, most importantly, background fiducialisation. This is particularly important for low-energy solar neutrino analyses, where the signal rate is low and the high energy tails of the background spectra leak into the signal region. The matching is also important for supernova neutrino analyses, where the time structure of the neutrino burst provides information about the supernova dynamics.

The matching efficiency grows moderately with particle energy and strongly with proximity to the PDS plane. In the case of the VD geometry, the efficiency also depends on the PDS plane being matched to, as the different planes have different PDS coverage (see **??**). The cathode plane has the highest efficiency, while the membrane planes closest to the anode have the lower efficiency. This is due to the fact that the light produced near the cathode plane has a higher chance of being detected by the PDS system, as it is closer to the LAr active volume.

Parameter	**(HD)**	**(VD)**
`AdjOpFlashX`	–	100 cm
`AdjOpFlashY`	100 cm	100 cm
`AdjOpFlashZ`	100 cm	100 cm
`AdjOpFlashMinNHitCut`	3	6
`AdjOpFlashMaxPERatioCut`	1	1
`AdjOpFlashMinPEAttenuate`	light_map	light_map
`AdjOpFlashMaxPEAttenuate`	light_map	light_map

Table 4: Flash-matching parameters for HD and VD geometries.

From this matching one can reconstruct the drift coordinate of the TPC cluster, which is essential for the correct reconstruction of the neutrino energy. The drift coordinate is computed from the time between the TPC vertex and the PDS flash by taking into account the drift velocity of the electrons in the LAr. Table 4 shows the clustering and matching parameters for each geometry used for this previous results. Note that the clustering parameters will be held the same for the different background configurations, while the matching parameters can be optimized for each configuration to maximize the matching efficiency.

For any given TPC cluster, the matching strategy follows these steps:

1. **Distance Calculation**: Compute the distance between the TPC cluster vertex and all PDS flash positions.

2. **Time Difference Calculation**: Calculate the time difference between the TPC cluster and the PDS flash. This must be smaller than the drift time of the electrons in the LAr.

3. **Candidate Selection Criteria**: A PDS flash is considered adjacent to a TPC flash if its projected distance follows:

$$\left(\frac{\mathrm{TPC_y} - \mathrm{PDS_y}}{\mathrm{AdjOpFlashY}}\right)^2 + \left(\frac{\mathrm{TPC_Z} - \mathrm{PDS_Z}}{\mathrm{AdjOpFlashZ}}\right)^2 < 1\,. \tag{6}$$

   Here AdjOpFlashY and AdjOpFlashZ are the adjusted matching distances in the 2 projected coordinates, respectively. Similarly, in the case of the VD geometry matching against membrane flashes, and additional AdjOpFlashX parameter is used where the $(\mathrm{TPC_X} - \mathrm{PDS_X})^2/\mathrm{AdjOpFlashX}^2$ term is computed from the time difference between the TPC cluster and the PDS flash.

4. **Matching Criteria**: The algorithm consider 3 additional cuts to prevent the random matching of background cluster:

   - The minimum number of hits in a PDS flash: `AdjOpFlashMinNHitCut` (by default 3).
   - The maximum ratio of `AdjOpFlashMaxPE` / `AdjOpFlashPE`: `AdjOpFlashMaxPERatioCut` (by default 1). This is intended to prevent the matching of background PDS flashes, which are less homogeneous in the number of PEs compared to the signal PDS flashes.
   - The minimum & maximum number of PEs `AdjOpFlashMinPECut` & `AdjOpFlashMaxPECut`. This cut is tuned to match the expected light-map of a neutrino production.

This algorithm reduces the initial selection of PDS flashes to a set of matching candidates. From this set, only one flash is selected as the match for the TPC cluster. To do so, a light-map based parametrization has been implemented by evaluating the expected number of PEs in the flash as a function of the distance to cluster's charge and predicted drift distance. Hence, the algorithm picks out the flash candidate that minimizes the following quantity:

$$\chi = \frac{\mathrm{abs}(\mathrm{PE_{flash}} - \mathrm{PE_{expected}})}{\mathrm{PE_{flash}^{\alpha}}}\,, \tag{7}$$

where $\mathrm{PE_{flash}}$ is the number of PEs in the flash, $\mathrm{PE_{expected}}$ is the expected number of PEs from the light-map parametrization, and α is a tunable parameter (by default set to 2) that controls the weight of the flash PE count in the matching.

This parametrization is helpful to investigate which matching strategy works best:

- **Minimum Error Matching** ($\alpha = 0$): select the flash candidate that minimizes the absolute difference between the number of PEs in the flash and the expected number of PEs from the light-map.
- **Minimum Relative Error Matching** ($\alpha = 1$): select the flash candidate that minimizes the relative difference between the number of PEs in the flash and the expected number of PEs from the light-map.

- **Weighted Relative Error Matching** ($\alpha = 2$): select the flash candidate that minimizes the weighted relative difference between the number of PEs in the flash and the expected number of PEs from the light-map.
- **Maximum PE Matching** ($\alpha \to \infty$): select the flash candidate with the highest number of PEs.

In the following we detail the implementation of the light-map assisted matching method. This algorithm takes into account the expected light yield distribution for a given signal TPC cluster:

$$\log_{10}(\mathrm{PE}) = A - A \cdot \lambda \cdot \frac{t}{t_{max}} + C \cdot \left(\frac{t}{t_{max}}\right)^2 . \tag{8}$$

Where A, λ and C are amplitude, attenuation, and correction fit parameters; t is the drift time and t_{max} is the maximum drift time in the detector. This formula can be used to extract different attenuation curves for different cluster charge ranges. Figure 14 shows the expected number of PEs in the flash as a function of the drift time for different cluster charge ranges for the HD and VD geometries. The extracted attenuation curve parameters can be fitted to simple quadratic functions in the logarithmic space to parametrize the light-map assisted PE cut as a function of both drift time and cluster charge.

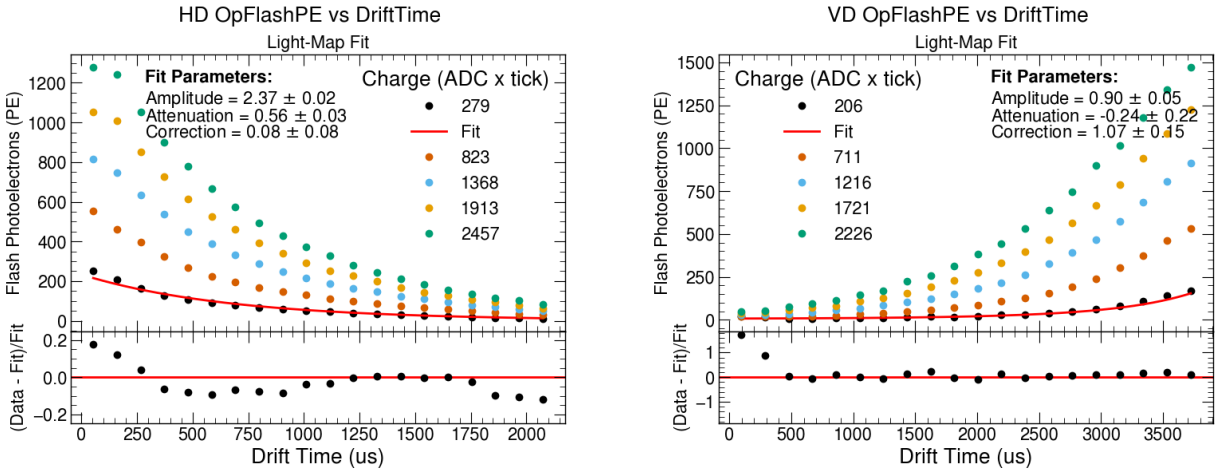


Figure 14: Light-map assisted flash-matching parametrization for the HD and VD geometries. The expected number of PEs in the flash is shown as a function of the drift time for different cluster charge ranges.

The extracted attenuation curve parameters can be fitted to simple quadratic functions in the logarithmic space to parametrize the light-map assisted PE cut as a function of both drift time and cluster charge Q:

$$\mathrm{PE}_{\mathrm{cut}}(t, Q) = \exp\left[A(Q) - A(Q) \cdot \lambda(Q) \cdot \frac{t}{t_{max}} + C(Q) \cdot \left(\frac{t}{t_{max}}\right)^2\right] , \tag{9}$$
$$\text{with } \alpha(Q) = a_0 + a_1 \cdot Q + a_2 \cdot Q^2 \quad \text{where } \alpha \in \{A, \lambda, C\} .$$

Where $A(Q)$, $\lambda(Q)$ and $C(Q)$ are amplitude, attenuation, and correction fit parameters as a function of the cluster charge Q; t is the drift time and t_{max} is the maximum drift time in the

detector. Figure 15 shows the extracted parameter curves as a function of the cluster's charge for the HD and VD geometries. As an example, the amplitude parameter is shown fitted to a quadratic function. The combination of all fitted parameters provides a parametrization to compute the expected number of PEs in the flash for a given cluster charge and drift time.

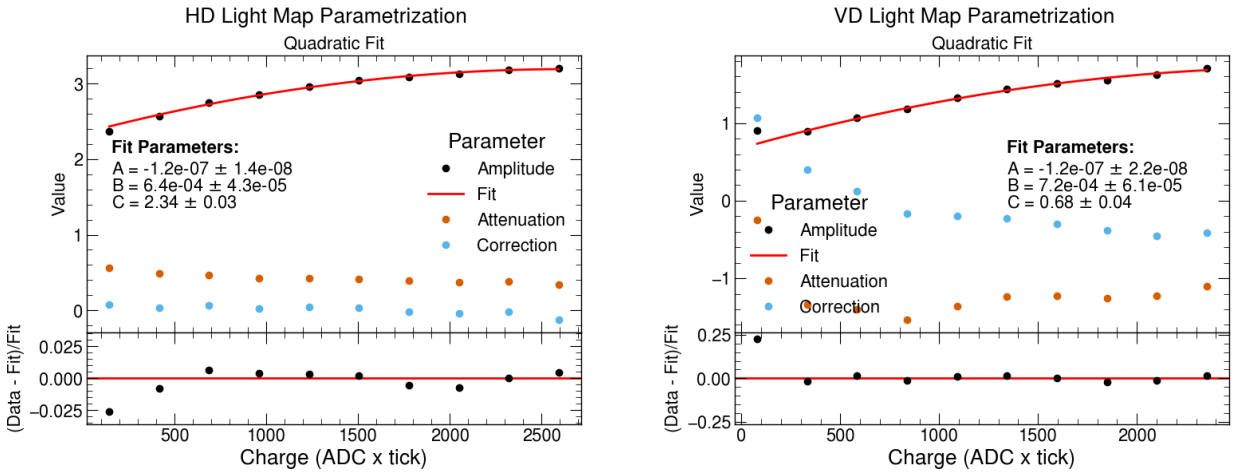


Figure 15: Light-map assisted flash-matching parametrization for the HD and VD geometries. The extracted attenuation curve parameters are shown as a function of the cluster charge.

In terms of flash-matching efficiency, we can see in table 5 that the HD geometry has a better overall performance than the VD geometry, which is expected due to the larger drift volume of the latter. The vd_1x8x14_3view_30deg_nominal configuration has the lower matching efficiency due to the larger drift volume and higher background rate, which increases the probability of miss-matching.

TPC–PDS Matched Neutrino Sample (%)

Configuration	No Match	Background	Low–Purity	High–Purity
HD Signal	1.72(2)	—	—	98.3(2)
HD Central	0.56(3)	39.0(2)	1.29(4)	59.1(3)
HD Lateral	0.56(3)	34.8(2)	0.44(2)	64.2(3)
VD Signal	12.32(7)	—	—	87.7(2)
VD Nominal	5.9(2)	54.5(7)	34.6(5)	5.0(2)

Table 5: TPC-PDS matching comparison between geometries. Fraction of sample matched to flash.

The matching efficiency is defined as the fraction of TPC clusters that have been matched to a signal PDS flash. Figure 16 show the flash-matching efficiency for the HD and VD signal productions as function of the different parameters. Note that these have been used as a figure of merit to optimize the matching parameters described above. The goal is to maximize the matching efficiency (>90%) while minimizing the reach of the clustering parameters to avoid random background matches.

One can see that the matching efficiency decreases with increasing drift distance, which is expected due to the attenuation of the light in the LAr. The HD geometry has a higher matching efficiency than the VD geometry, especially at larger drift distances. This is expected due to the larger drift volume of the VD geometry, which leads to a higher probability of miss-matching.

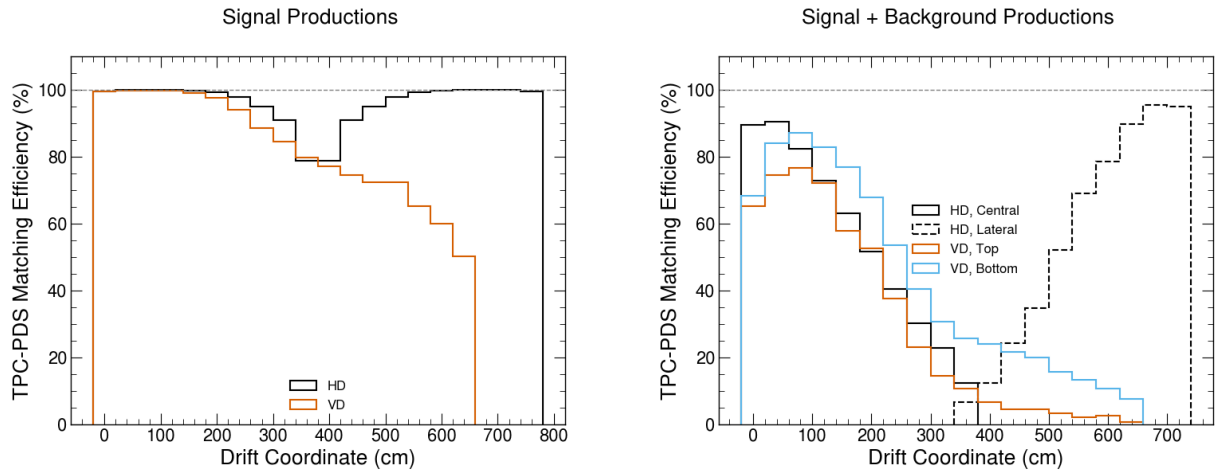


Figure 16: Flash-matching efficiency for the HD and VD signal productions as function of the drift coordinate. (Left) Matching efficiency for signal productions. (Right) Matching efficiency for signal and background productions.

3.4 Event Definition

We define a low-energy event as the combination of a `primary cluster` together with its `adjacent clusters` and a list of `flashes` that are matching candidates for the drift coordinate reconstruction (see section 3.3). In any neutrino analysis it will become important to try to distinguish (at least on average) the `adjacent cluster` distributions for any low-energy event, not only to distinguish background activity (e.g. neutron captures) from signal events, but also to identify the different interaction channels of the neutrino interaction (CC events produce de-excitation gammas that are not present in ES interactions). Consequently, the higher the projected background activity in any of the isolated drift volumes of DUNE's FDs, the harder particle identification and channel separation will become. Notice that there is nothing preventing a single neutrino interaction from producing more than one low-energy event. Rarely, but not inevitably, secondary gammas scatter further than 1 m in LAr or a background-induced miss-match in the TPC reconstruction leads to blips appearing in uncorrelated regions of the detector. Independently of this possibility, a low-energy event reconstruction efficiency can be defined as the possibility of a single particle interaction to produce at least one low-energy event that can be fully reconstructed. This efficiency is shown in fig. 17 for both HD and VD geometries as a function of the cluster's `nhit` threshold value. The chosen signal clustering parameters are 3 adjacent channels and 12.5 µs time window. The results show a clear and expected decrease in efficiency with increasing `nhit` threshold, as low-energy events are more likely to be lost when applying stricter cuts on the minimum number of hits per cluster. This shows the importance of a simple clustering approach to low-energy neutrino reconstruction, as more sophisticated algorithms might lead to a considerable decrease in efficiency.

The first figure of merit to be extracted from the simulation is the detection efficiency (see fig. 17). We observe a strong dependence on the number of hits. Achieving a reconstruction threshold (>90% efficiency) above 5.5 MeV requires a minimum hit condition of 1 hit.

The next parameter to evaluate is the completeness of the main clusters, which is a measure of the electron charge–confinement given a certain clustering parameter set. For the purpose of the solar neutrino analysis of DUNE, clustering parameters of 3 adjacent channels and 25 ticks have been used to maximize completeness while ensuring cluster purity in the context of background. The most up-to-date values for this clustering should be found in the producer's

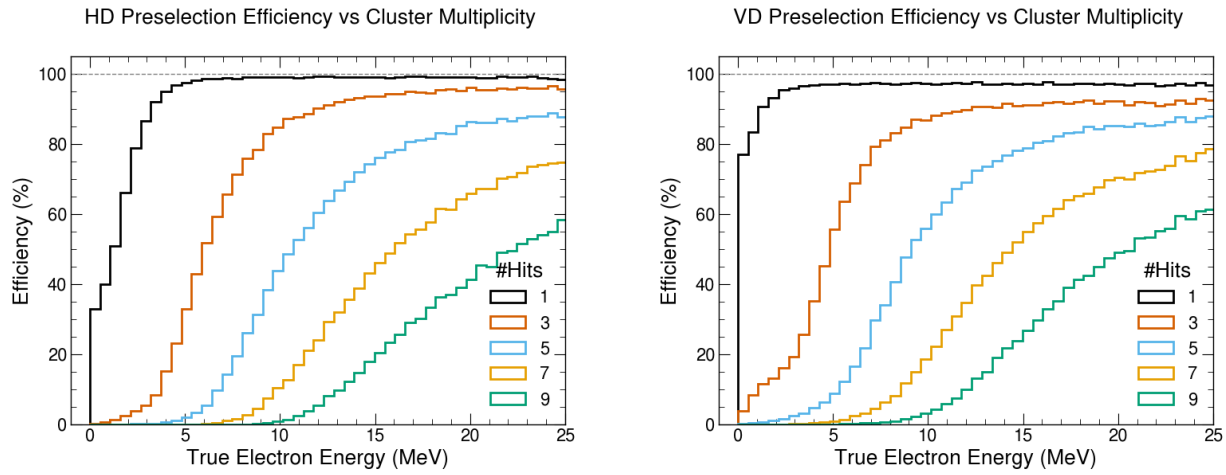


Figure 17: HD (left) and VD (right) detection efficiency for signal clustering parameters of 3 adjacent channels and 12.5 µs.

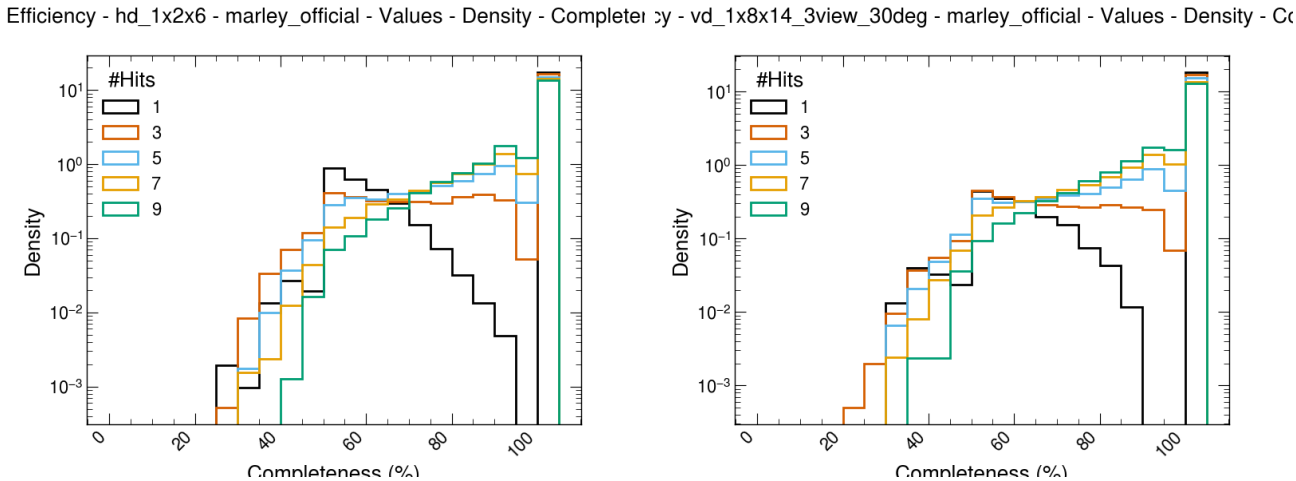


Figure 18: Detection efficiency for the HD (left) and VD (right) geometries. Results are shown as a function of the minimum number of hits per cluster and true neutrino energy.

SolarCluster.fcl. Despite the good overall completeness of the signal, we observe a trend with respect to the number of hits per cluster, which can be accounted for in the later described charged correction factor to improve reconstruction. The trend that one observes is clear and understandable; for depositions which result in a lower number of hits (this can happen because of low energy deposition or a perpendicular trajectory of the particle with respect to the TPC plane) it is more likely that clusters of similar sizes are formed (e.g. 2 clusters of 1 hit) resulting in a lower charge in the main cluster.

DUNE's solar analysis workflow starts with calibrating the clustered hits from the CC interaction of simulated neutrino events with the MARLEY generator. The CC channel is chosen to be the default based on its dominant cross-section in the low-energy range, as well as its richer topology. All energy and spacial resolution results presented here should be considered an underestimation (or at least compatible) to the ones results for the ES channel. Each cluster's information is reduced to a point-like object characterized by its number of hits (`nhits`), `charge` (summed over the charge of the individual hits) and a 2.5D-reconstructed vertex $< t, y, z >$ computed from the charge-weighted average of the individual hits.

For the purpose of this work, secondary low-energy events coming from a single neutrino inter-

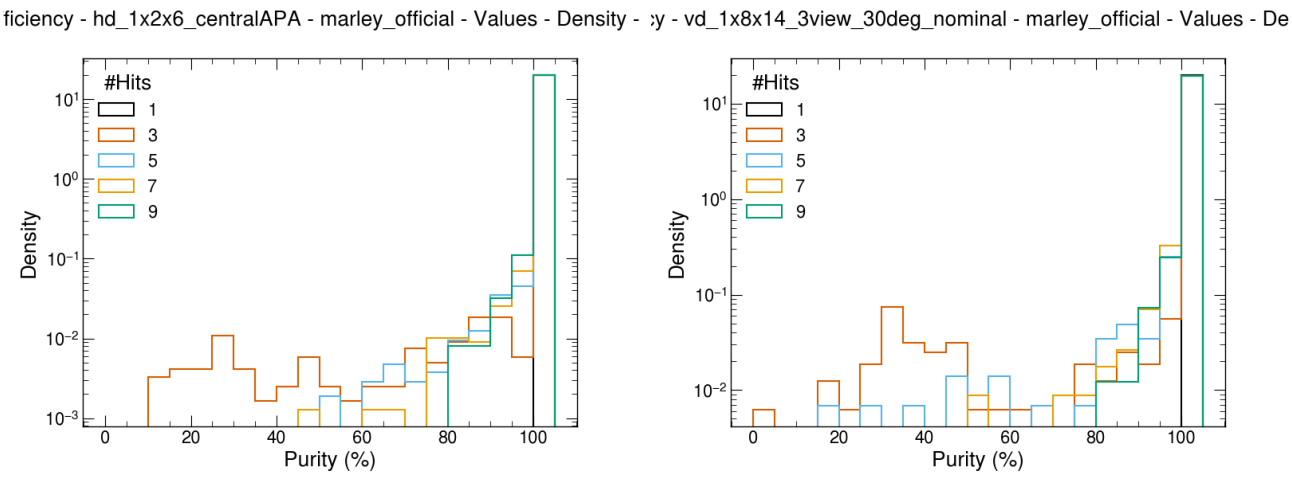


Figure 19: Cluster purity for the HD (left) and VD (right) geometries. Results are shown as a function of the minimum number of hits per cluster and true neutrino energy.

Adjacent Cluster Distribution - Vertex Distance 100 cm

Configuration	Source		
	Signal	Intrinsic	External
HD Signal	1.277 ± 0.003	—	—
HD Lateral	1.048 ± 0.003	0.2094 ± 0.0010	0.311 ± 0.001
HD Central	0.941 ± 0.003	0.234 ± 0.001	0.1739 ± 0.0010
VD Signal	1.489 ± 0.004	—	—
VD Nominal	0.988 ± 0.007	2.37 ± 0.01	3.43 ± 0.02
VD Shielded	1.017 ± 0.006	2.30 ± 0.01	0.778 ± 0.005

Table 6: Average number of adjacent clusters per primary signal cluster for 100 cm distance.

action will always be ignored in favor of the one with the highest charge in the primary cluster. This is to ensure that the reconstruction performance is evaluated for the most complete reconstruction object in any interaction but in any physics analysis these should be kept and treated as part of the signal or even background samples.

Adjacent Cluster Distribution - Vertex Distance 20 cm

Configuration	Source		
	Signal	Intrinsic	External
HD Signal	0.759 ± 0.002	—	—
HD Lateral	0.609 ± 0.002	0.0165 ± 0.0003	0.0228 ± 0.0003
HD Central	0.538 ± 0.002	0.0181 ± 0.0002	0.0131 ± 0.0002
VD Signal	0.890 ± 0.002	—	—
VD Nominal	0.503 ± 0.004	0.147 ± 0.002	0.224 ± 0.002
VD Shielded	0.531 ± 0.004	0.145 ± 0.002	0.0518 ± 0.0009

Table 7: Average number of adjacent clusters per primary signal cluster for 20 cm distance.

For the specific purpose of the solar neutrino studies an analyzer module (SolarNuAna_module.cc) has been written to combine these reconstruction objects with the truth information of the event. This module essentially combines all the information needed to perform the calibration and reconstruction of the signal candidates. The biggest caveat from the analysis point of view is that

all clusters and flash information is reduced to a point-like data structure. This is, of course, a choice of design in the analyzer and allows for a simpler handling of the data because all event candidates have the same dimension. In future works, this analyzer should be substituted by the more general LowEAna_module.cc which aims to provide the same event reconstruction while keeping the internal structure of the candidate clusters to allow for better calibration and discrimination strategies.

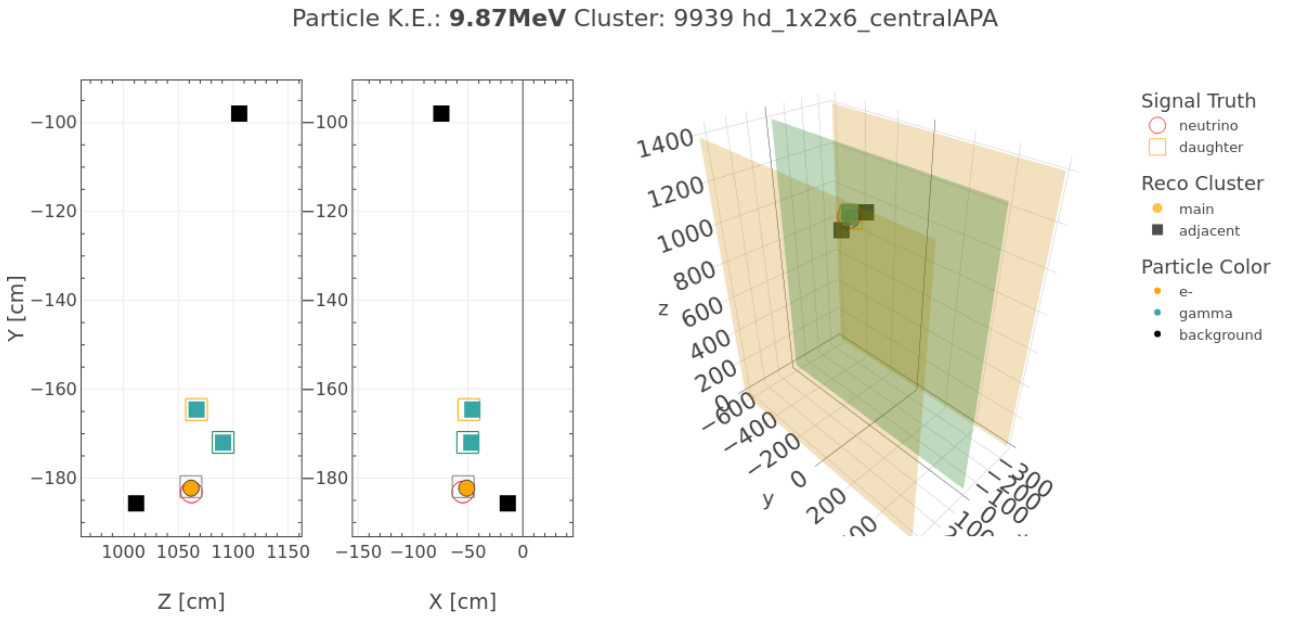


Figure 20: Example of a simulated ∼10 MeV neutrino interaction in the HD geometry. (left) The event displays the two TPC projections of the primary electron cluster (orange), the associated de-excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. (right) The same points are displayed in a 3D representation of the workspace geometry. The represented planes are the central APA (green) and the corresponding CPAs (orange)

An example of a simulated neutrino interaction in the HD geometry is shown in fig. 20. The event displays the two TPC projections of the primary electron cluster (orange), the associated de-excitation gammas (turquoise) and uncorrelated background activity (black). The biggest energy deposits from the truth information of the signal propagation are marked with empty squares and the reconstructed clusters with filled squares. Compared to fig. 3, the clusters are now reduced to a point-like data structure, which results from the clustering algorithm.

4 Neutrino Energy Calibration

The calibration and characterization of the CC neutrino signal are based on the true information of the main electron of the interaction. If not otherwise mentioned, the plots in this section are produced from a MARLEY generated CC neutrino sample with central APA background. Figure 21 shows the correlation between the neutrino's main electron energy and the number of hits in the main TPC cluster, which is used as a proxy for the electron energy computation in the calibration process.

The first step of the reconstruction process has already happened when clustering the hits of the

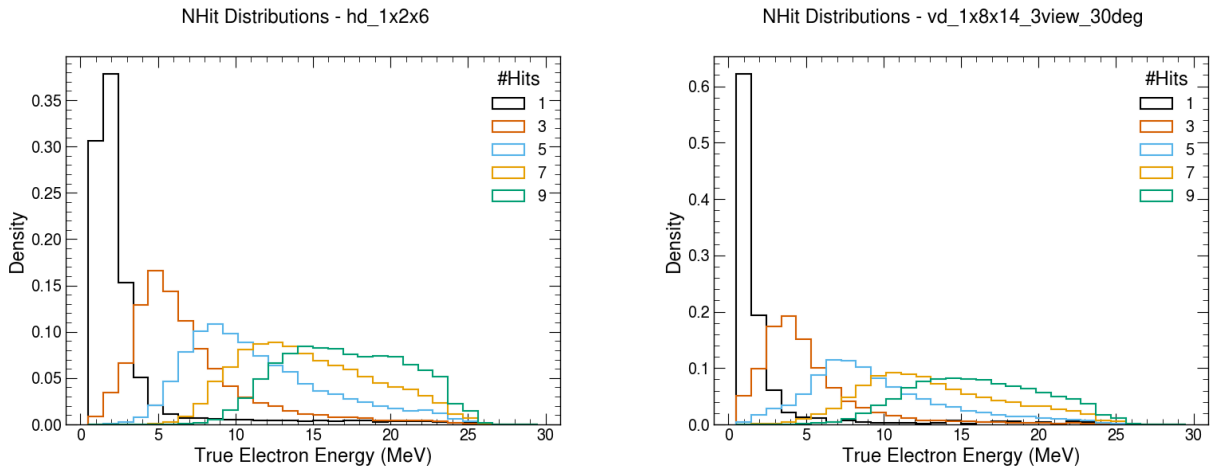


Figure 21: Neutrino energy correlation with number of hits in the primary cluster.

TPC and PDS detector subsystems in the simulation's *reco1* stage. Subsequently, the electron's charge is corrected for the drift attenuation computing a hit-dependent calibration factor that recover the particle's original kinetic energy (see sections 4.1 and 4.2). Finally, the average calibration factor is applied to all adjacent clusters of the event candidate and the total neutrino energy is computed from four different algorithms (see sections 4.3 to 4.5). During the TPC calibration process, the drift coordinate is assumed from truth. A discussion on how the intrinsic background contamination of the detector affects the signal reconstruction and the effects on drift coordinate resolution (as a result of TPC-PDS match) are presented together with the vertex and energy resolution results in section 5.

4.1 Electron-Lifetime Correction

TPC hits are first grouped into spatial clusters using a dedicated clustering algorithm. Each event contains a **main cluster**, corresponding to the primary outgoing electron, and zero or more **adjacent clusters**, interpreted as gamma-induced "blips".

In order to calibrate the charge attributed to each of these clusters, the first conceptual step is to understand that there is a direct correlation between the particle's energy and the collected charge. Keeping aside the differences in electron yield that each kind of particle produces and focusing on the main electron of the CC $\nu - \mathrm{Ar}$ interaction, we identify the main cluster as the one with the highest charge and assume that it corresponds to a fully-contained electron. This assumption allows us to use the known true energy of the main electron (from simulation) as a reference for calibrating the charge-to-energy conversion factor:

$$Q^{\mathrm{coll}}(n_{\mathrm{hit}}) = f\left(E_e^{\mathrm{true}}, \overrightarrow{p}, n_{\mathrm{hit}}\right) \cdot E_e^{\mathrm{true}}. \tag{10}$$

Here, Q^{coll} is the charge collected in the main cluster, E_e^{true} is the true energy of the main electron, and f is the theoretical charge-to-energy conversion factor that depends on the true energy, the vertex of the electron's interaction (represented by $\overrightarrow{p}$), and the number of hits in the resulting cluster (n_{hit}).

From an experimental perspective, the charge collected in the main cluster is affected by various factors, that are a priory unknown, including the particles energy and position. We define an

effective charge-to-energy conversion factor, $F_0(n_{\text{hit}})$, that encapsulates all these effects as a function of the number of hits in the cluster:

$$F_0(n_{\text{hit}}) = \frac{1}{\Delta t}\int_0^{\Delta t} \frac{Q^{\text{corr}}(t\,, n_{\text{hit}})}{E_e^{\text{true}}} dt \quad \text{with} \quad Q^{\text{corr}}(n_{\text{hit}}) = \frac{Q^{\text{coll}}(n_{\text{hit}})}{e^{-\frac{t}{\tau_e}}}\,. \tag{11}$$

$$E_e^{\text{cluster}} = F_0(n_{\text{hit}}) \cdot Q^{\text{corr}}(n_{\text{hit}})\,. \tag{12}$$

Here Q^{corr} is the lifetime-corrected cluster charge, E_e^{cluster} is the computed electron energy for the cluster, Δt is the maximum drift time of the electron, and τ_e is the electron lifetime measured here independently and cross-checked against the simulation input.

The first step in calibration involves converting the total collected charge of each cluster to a computed electron energy by correcting for the electron-lifetime attenuation in the LAr (see section 4.1). Figure 22 shows the result of this procedure.

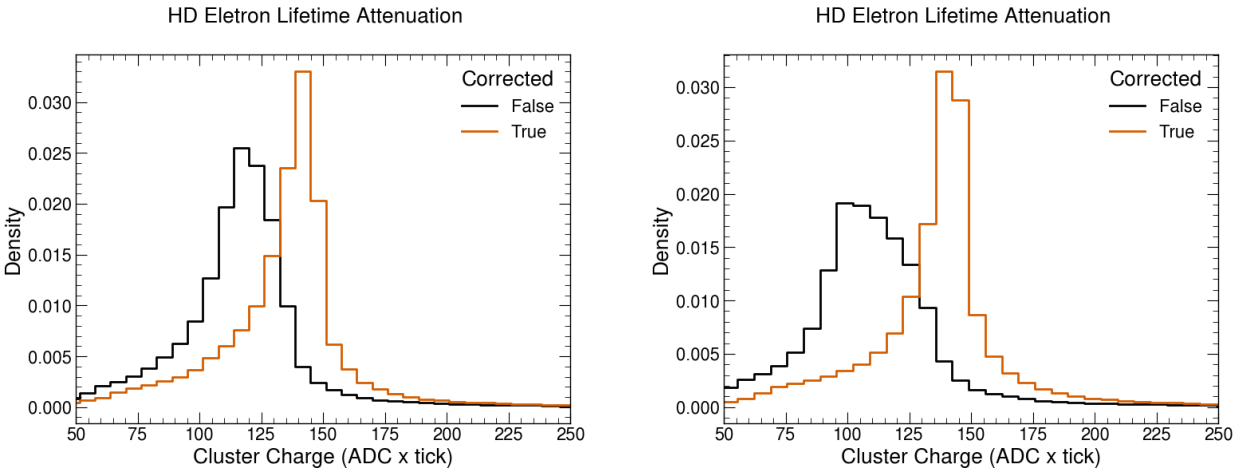


Figure 22: Drift attenuation correction for HD (left) and VD (right).

Figure 23 shows the charge convertion factor as a function of the clusters n_{hit}. Due to the nature of clustering, the first points of the distribution are affected by threshold effects, which can be read from the plot by analysing the large error bars. Intermediate n_{hit} clusters have the lowest charge-to-energy ratio, as they are more likely to be fully contained. With increasing n_{hit}, the comp‘leteness decreases, leading to a higher charge-to-energy ratio. To assess the impact of this calibration, with the goal of characterizing signal clusters in the context of background with limited the clustering efficiency, one can also investigate a “cheated charge”. This is computed for all clusters, adding the surrounding charge that has been backtracked to the main electron of the interaction. A calibration based on this cheated clustering gives us an understanding of the losses that arise from a limited clustering ability.

This trend is well-described by an exponential plus logistic function. The functional form used to fit the charge correction factor as a function of the number of hits is given by:

$$F_0(n_{\text{hit}}) = a \cdot e^{-b \cdot n_{\text{hit}}} + \frac{c}{1 + e^{-d \cdot n_{\text{hit}}}}\,, \tag{13}$$

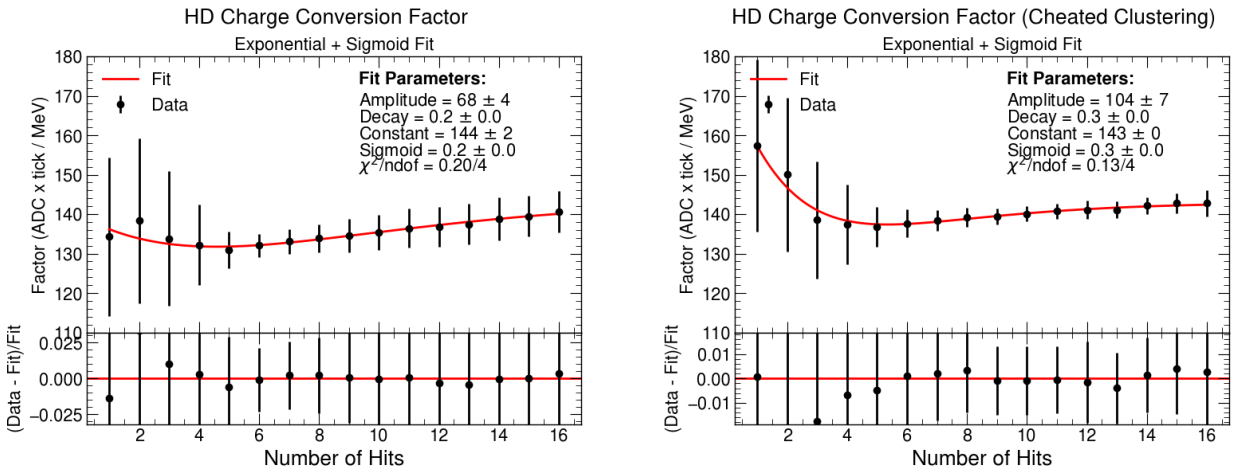


Figure 23: Per hit correction factor distributions. (left) Primary cluster charge. (right) Cheated electron cluster charge.

where a, b, c, d are fit's exponential amplitude, decay, limiting constant, and sigmoid growth parameters, respectively. This function captures the initial rapid decrease in the correction factor for low hit counts, followed by a more gradual increase at higher hit counts, reflecting the underlying physics of charge collection and clustering in the detector.

4.2 Electron Energy Calibration

Even after lifetime and charge calibration, small systematic biases may remain in the reconstructed electron energy due to recombination modelling, imperfect response, or clustering inefficiencies. We apply a linear correction to account for this residual offset:

$$E_e^{\text{cluster}} = \alpha\left(E_e^{\text{true}},\ \overrightarrow{p}\ , n_{\text{hit}}\right) \cdot E_e^{\text{true}} + \beta\left(E_e^{\text{true}},\ \overrightarrow{p}\ , n_{\text{hit}}\right) \tag{14}$$

Here, E_e^{cluster} is the computed electron energy for the cluster, E_e^{true} is the true energy of the main electron, and α and β are the linear correction parameters that depend on the true energy, the vertex of the electron's interaction (represented by $\overrightarrow{p}$), and the number of hits in the resulting cluster (n_{hit}). These parameters are determined by fitting the distribution of E_e^{cluster} versus E_e^{true} for the main clusters across the dataset.

As shown in fig. 25, the resulting calibration parameters approach unity for a and continuously rise for b as the number of hits increases, indicating worse electron clustering for higher-energy particles. There is no significant difference between the HD and VD geometries in terms of these calibration parameters.

The final calibrated electron energy, E_e^{reco}, is obtained by applying the inverse of the linear correction to the cluster energy, effectively removing the bias and improving the agreement between reconstructed and true energies, as illustrated in fig. 24.

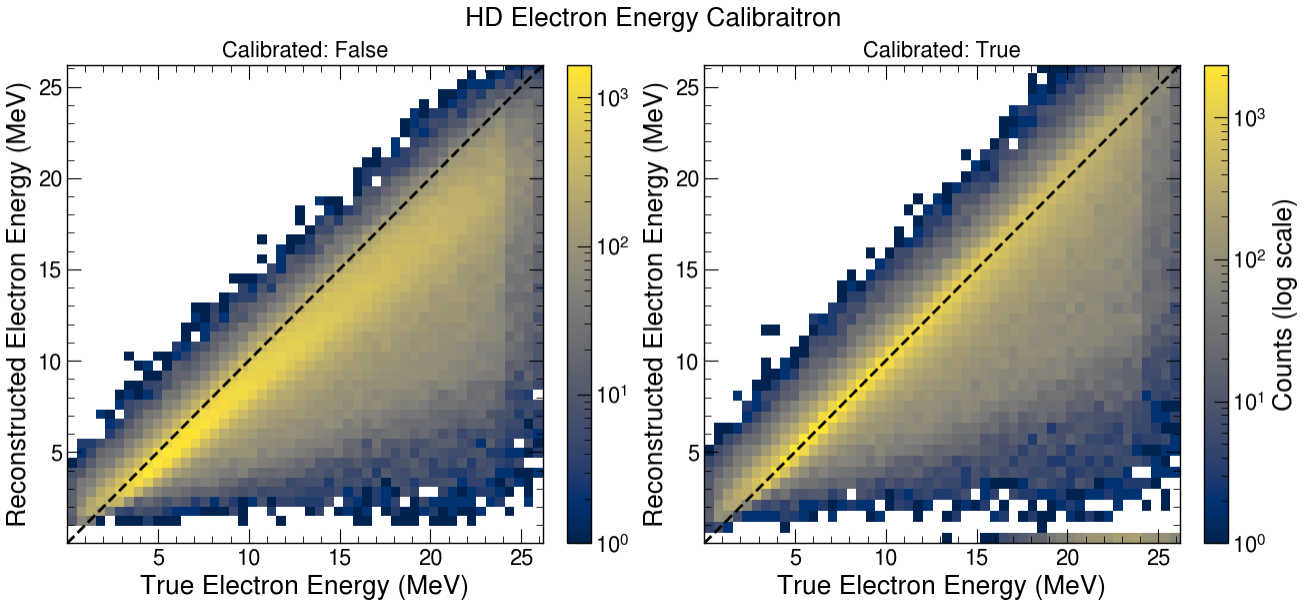


Figure 24: Charge calibration for the main electron clusters. (left) Comparison between true and reconstructed energy before linear fit calibration. (right) Comparison between true and reconstructed energy after linear fit calibration. Dashed line shows the values for which $E_{\text{reco}} = E_{\text{true}}$ as reference.

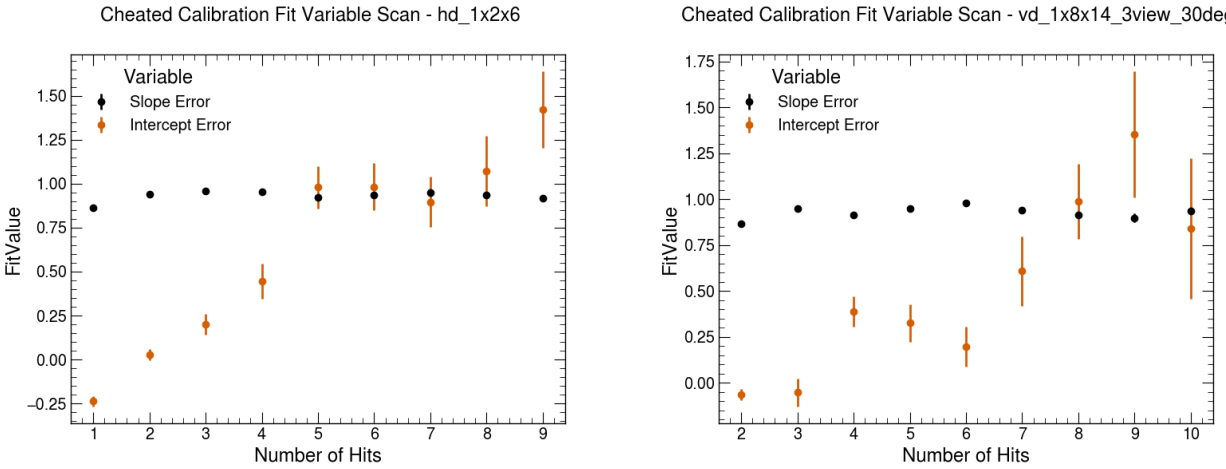


Figure 25: Calibration parameters as a function of number of hits in the cluster for (left) HD and (right) VD.

$$E_e^{\text{reco}} = \frac{E_e^{\text{cluster}} - b(n_{\text{hit}})}{a(n_{\text{hit}})} \tag{15}$$

$$\text{with} \quad \{a(n_{\text{hit}}),\, b(n_{\text{hit}})\} = \mathcal{A}\left[\alpha\left(E_e^{\text{true}},\, \overrightarrow{p}\,, n_{\text{hit}}\right),\, \beta\left(E_e^{\text{true}},\, \overrightarrow{p}\,, n_{\text{hit}}\right)\right] .$$

?? shows the final step of the electron energy calibration, where $\mathcal{A}$ represents the fitting procedure used to determine the parameters $a(n_{\text{hit}})$ and $b(n_{\text{hit}})$ from the distributions of α and β as functions of the number of hits in the cluster. This calibration ensures that the reconstructed electron energy is as close as possible to the true energy, accounting for any systematic biases that may arise from detector effects or reconstruction inefficiencies.

4.3 Adjacent Cluster Calibration

Applying this calibration to the main electron cluster, we obtain the final calibrated energy of the main electron. For all the adjacent clusters, we simply apply the average charge correction factor obtained from the main electron calibration.

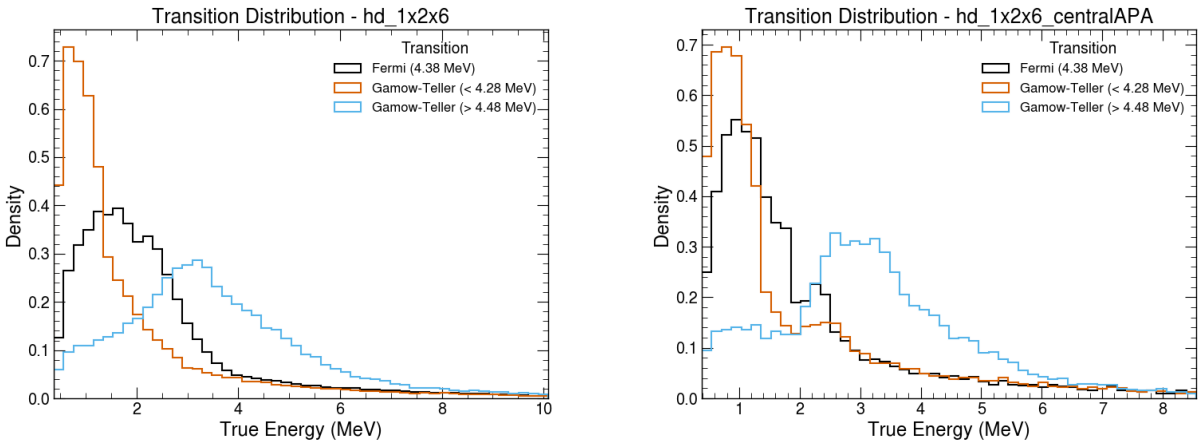


Figure 26: Adjacent cluster signal (left) compared to the selected adjacent clusters (right) for the HD productions with and without background activity.

We observe a bias in the reconstructed gamma energy (>2 MeV), which is due to the clustering inefficiencies and the low energy of the gammas. This bias leads to a poor identification of the nuclear transitions in the neutrino energy spectrum. In fig. 26, we show the reconstructed gamma energy distributions for the different nuclear transition types: Gamow-Teller and Fermi. The Fermi transition has a single emission line at 4.38 MeV, while the Gamow-Teller has several and its distribution is therefore divided into two regions: the low-energy region (<4.38 MeV) and the high-energy region (>4.38 MeV).

The bias in the reconstructed gamma energy could be corrected by applying a linear calibration, similar to the one applied to the main electron cluster. However, due to the additional presence of background, it has been considered to focus on improving the channel selection and clustering efficiency without further bias. Additional work needs to be performed in order to estimate the reconstruction potential that can be obtained from correctly tagging these nuclear transitions.

When considering the presence of background activity, as shown in fig. 26, the identification of the nuclear transitions becomes even more challenging. The background activity introduces additional low-energy deposits that can be misidentified as gammas from the neutrino interaction, further smearing the reconstructed gamma energy spectrum. This highlights the importance of effective background mitigation strategies and advanced reconstruction techniques to accurately identify and reconstruct the gamma energies associated with neutrino interactions in a realistic detector environment.

Another way to illustrate the challenge of reconstructing the gamma energies in the presence of background is to look at the true energy distributions of the adjacent clusters for different background scenarios. Figure 27 shows how the presence of background activity significantly increases the number of low-energy adjacent clusters (<3 MeV), which can be misidentified as gammas from the neutrino interaction. From this comparison it is clear that the shielded background scenario reduces considerably the amount of low–energy deposits, but this effect becomes weaker at higher energies.

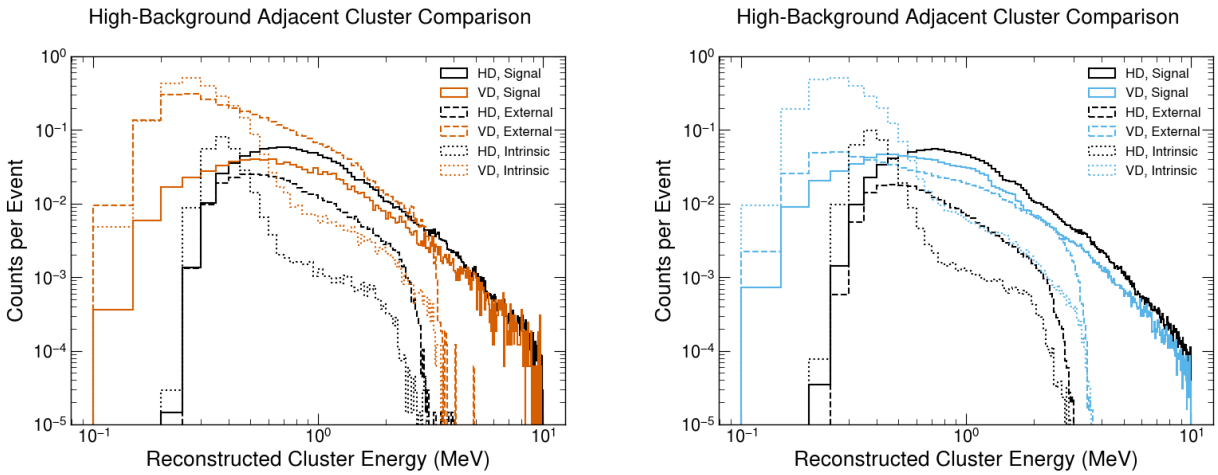


Figure 27: Adjacent cluster distributions for signal and background samples. (left) High-background geometries (hd_1x2x6_lateralAPA & vd_1x8x14_3view_30deg_nominal). (right) Low-background geometries (hd_1x2x6_centralAPA & vd_1x8x14_3view_30deg_shielded).

4.4 Neutrino Energy Reconstruction

In this section we describe the different methods used to reconstruct the neutrino energy from the main electron cluster and the adjacent clusters. We present four reconstruction algorithms: **TotalEnergy**, **SelectedEnergy**, **SolarEnergy** and **ClusterEnergy** which will be compared in terms of energy resolution in section 5.5.

The first attempt in computing the neutrino energy is to add the main electron's lifetime-corrected and the charge calibrated energy (as described in section 4.2) to the energies of all adjacent clusters (up to 1 m). The adjacent clusters are assumed to be gamma-induced "blips" and are calibrated using the same charge calibration factor as the main electron. The total neutrino energy is computed as:

$$E_{\nu}^{\text{reco}} = E_{e}^{\text{reco}} + \sum_{i} E_{\text{adj, i}}^{\text{reco}} \,. \tag{16}$$

where E_{e}^{reco} is the reconstructed energy of the main electron cluster after charge calibration, and $E_{\text{adj, i}}^{\text{reco}}$ are the reconstructed energies of the adjacent clusters, also after charge calibration.

This strategy is the most obvious approach for neutrino energy reconstruction. We denote it as the **TotalEnergy** algorithm. However, it does not account for the energy smearing of the signal in the presence of background. Therefore, we study alternative methods to improve the energy resolution and mitigate the effects of background contamination.

The next solution is to apply a selection on the adjacent clusters (**SelectedEnergy** algorithm), which are assumed to be gamma-induced "blips". The selection is based on the adjacent cluster charge and distance to the main electron cluster. The adjacent clusters are selected if they have a charge above a certain threshold and are within a certain distance to the main electron cluster. The thresholds are tuned to maximize the signal over background ratio. fig. 28 shows the distribution of the adjacent clusters charge and distance to the main electron cluster and selection thresholds.

Table 8 summarizes the different thresholds used for the adjacent cluster selection, based on the

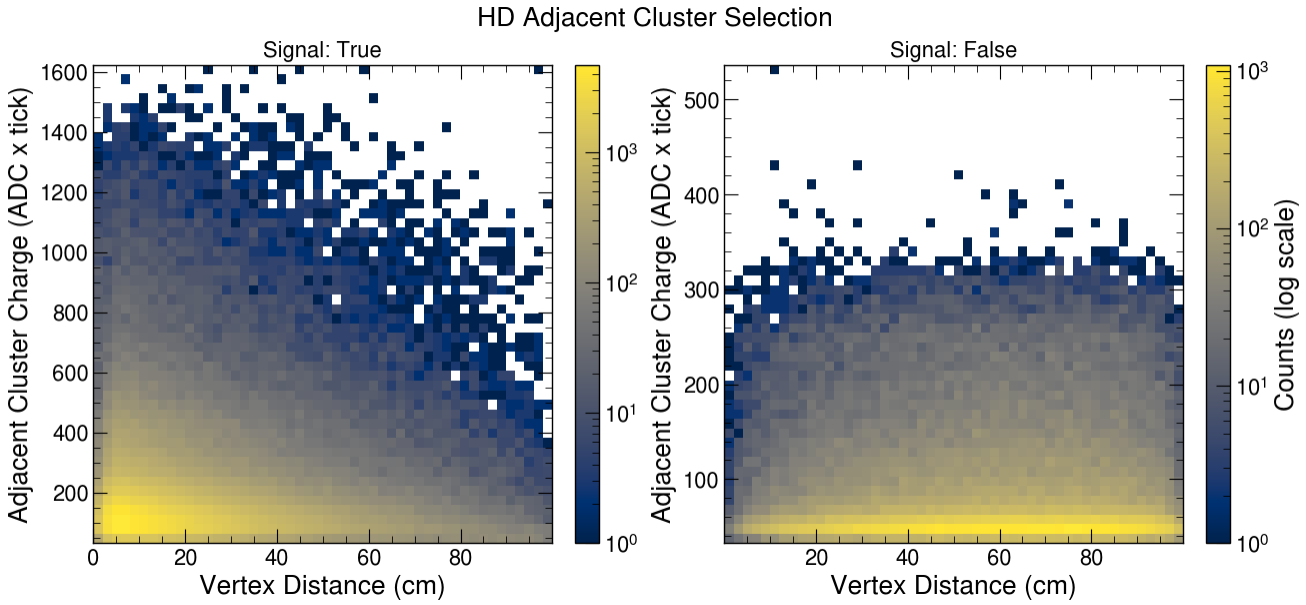


Figure 28: Distribution of the adjacent clusters charge and distance to the main electron cluster.

Configuration	Charge	Distance
hd_1x2x6_centralAPA	275 ADC tick	12.1 cm
hd_1x2x6_lateralAPA	309 ADC tick	11.0 cm
vd_1x8x14_3view_30deg_nominal	304 ADC tick	13.6 cm
vd_1x8x14_3view_30deg_shielded	308 ADC tick	17.6 cm

Table 8: Adjacent cluster selection thresholds for the different background scenarios.

background scenario. The selected adjacent clusters are then summed to the primary cluster energy to obtain the reconstructed neutrino energy (**SelectedEnergy**).

A more advanced version of this last algorithm is to apply machine learning methods to the adjacent clusters. The **SolarEnergy** algorithm is trained on the main cluster properties and the average adjacent cluster features two add one of two total adjacent cluster energy values: 4.47 MeV and 6.41 MeV. The algorithm uses a random forest regressor, which predicts the effective energy gone to the adjacent clusters, which if subsequently added to the main electron energy to obtain the result.

Hereby, the problem is simplified to a binary choice. This is motivated by the two main calibration lines from the CC interaction displayed in fig. 6. The score of the random forest regressor, which provides an AUC of 0.77 for the test sample, means that the algorithm is able to distinguish between the two samples with a good efficiency. The random forest regressor is trained on the following features:

- NHits: number of hits in the main electron cluster.
- AdjClNum: number of total adjacent clusters.
- SelectedAdjClNum: number of selected adjacent clusters.
- Energy: energy of the main clusters.
- **TotalAdjClEnergy**: energy of the total adjacent clusters.

- SelectedAdjClEnergy: energy of the selected adjacent clusters.
- SelectedAdjClEnergyRatio: energy ratio of the selected to the total adjacent clusters.
- **MaxAdjClEnergy**: maximum energy of the total adjacent clusters.
- SelectedMaxAdjClEnergy: maximum energy of the selected adjacent clusters.
- TotalEnergy: reconstructed **TotalEnergy**.
- SelectedEnergy: reconstructed **SelectedEnergy**.

From which the highlighted features are the most important for the random forest regressor, as shown in fig. 29.

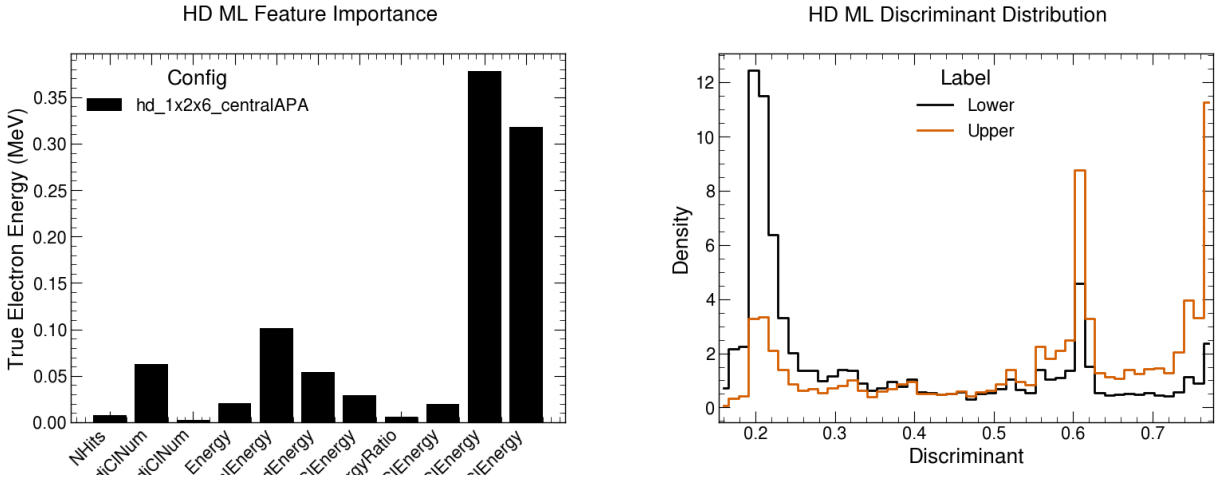


Figure 29: (left) Feature importance of the random forest regressor. (right) Discriminant calibration for the random forest regressor.

Finally, as a benchmark, we introduce the **ClusterEnergy** algorithm as the reconstruction method based purely on the main electron cluster. The only computation is the calibration step explained in the next section wich applied to the full sample.

4.5 Neutrino Energy Calibration

Due to the missing nuclear binding energy and undetected soft particles, the reconstructed neutrino energy is expected to systematically underestimate the true value. As a final step to the reconstruction process, all energies (**TotalEnergy**, **SelectedEnergy**, **SolarEnergy** and **ClusterEnergy**) are calibrated with a linear fit to compensate for systematic losses in the previous clustering or calibration processes. This final global correction is applied to bring the distribution into agreement with the simulation truth following a process equivalent to the one described in section 4.2 for the main electron cluster. The resulting neutrino energy smearing is shown in fig. 30.

The distribution is centered around the diagonal, indicating that the reconstructed energy is in good agreement with the true energy. The off-diagonal bands correspond to events where the reconstructed energy is significantly different from the true energy, which can be attributed to various factors such as undetected particles, misreconstruction, or fluctuations in the energy deposition. The overall shape of the distribution provides insight into the resolution and bias of the neutrino energy reconstruction, which is crucial for accurate physics analyses.

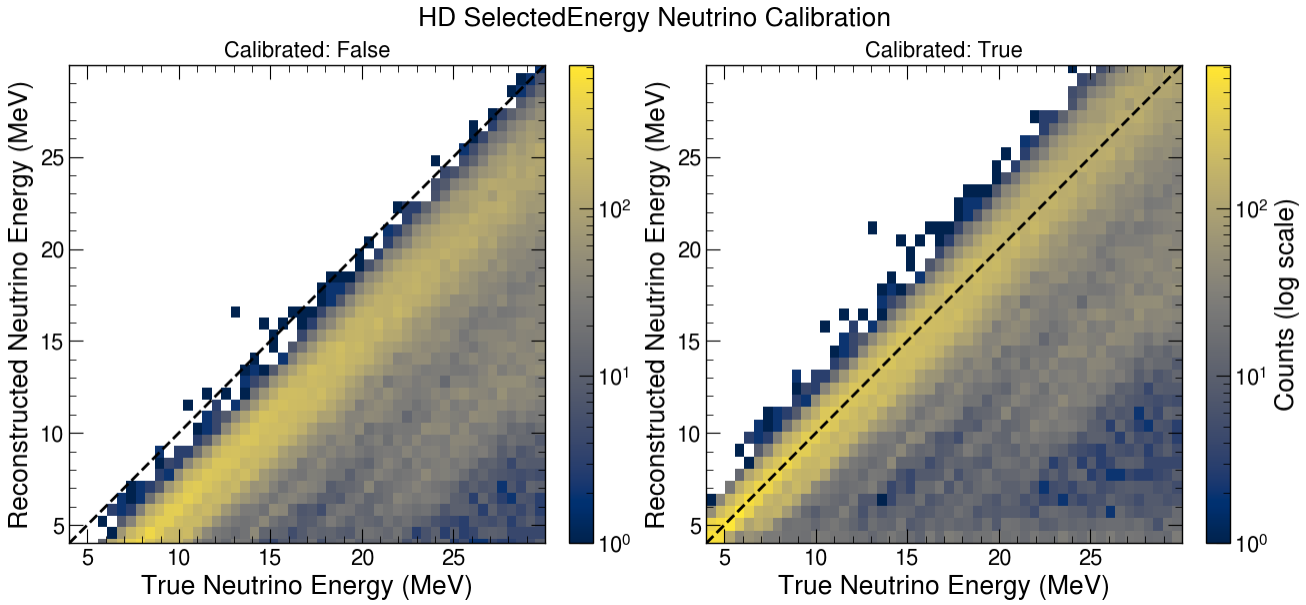


Figure 30: HD Neutrino energy smearing from the calibrated *SelectedEnergy* signal.

5 Results

From the TPC cluster vertex reconstruction and the flash–matching, we can extract the energy and vertex resolution. The calibrated neutrino energy is obtained from a standalone code SOLAR package, which formats the simulation output into numpy arrays and performs the successive operations presented in section 3. Here we present the main figures of merit and compare the resolution performance between the two geometries, the *hd_1x2x6_centralAPA* and the *vd_1x8x15_3view_30deg* and additional background reduced scenarios.

5.1 Neutrino Vertex Smearing

As explained in section 3, the drift-coordinate of the reconstructed vertex is obtained from the flash-matching algorithm, which combines the information from the TPC cluster and the associated PDS flash.

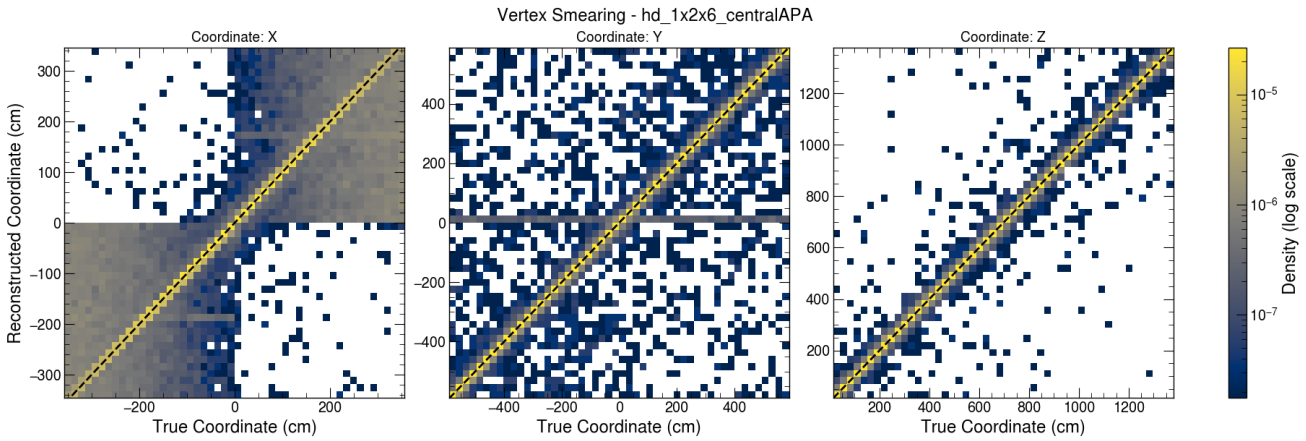


Figure 31: Vertex smearing from calibrated signal.

The results are shown in fig. 31 for the *vd_1x8x14_3view_30deg* geometry with nominal back-

ground production. These distributions are not Gaussian, as expected from the nature of the matching algorithms. The resulting matching can be correct, leading to a small error, or completely wrong, leading to a large error in the reconstructed coordinate.

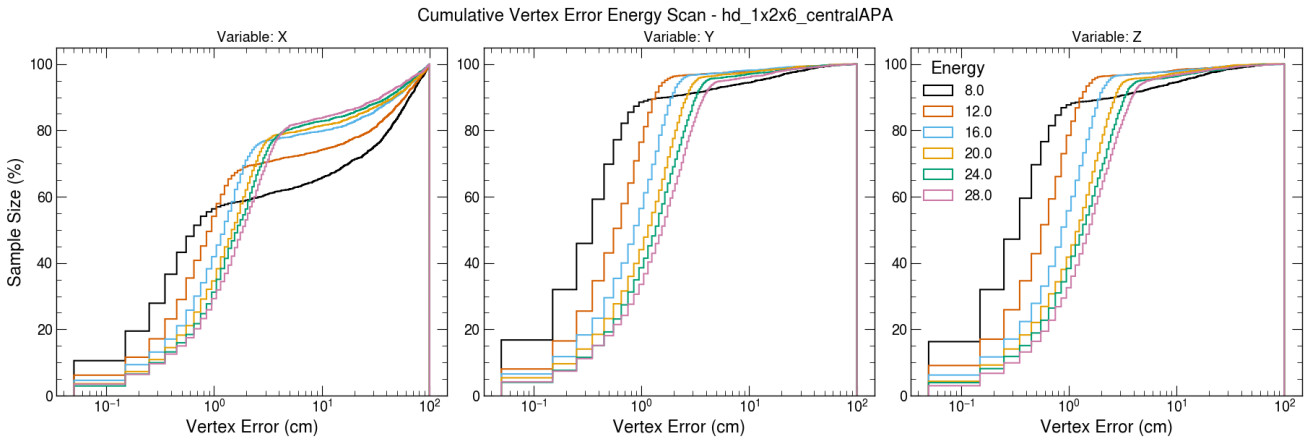


Figure 32: HD cumulative vertex error distribution. Showing x (left), y (center) and z (right) coordinates from left to right.

To have a better understanding of the reconstruction performance, we can compute the cumulative vertex error fig. 32. This plot shows the percentage of events that are reconstructed within a certain vertex error tolerance. For instance, we can see that approximately 90% of the events have a vertex error below 1 cm in the y- and z-coordinates, while only about 60% of the events achieve this level of accuracy in the x-coordinate.

5.2 Neutrino Vertex Resolution

The vertex resolution is defined from the distance between the reconstructed vertex and the true vertex position. Note that this analysis reconstructs particle interactions as point–like objects and therefore reconstructs a vertex that is shifted on average to the center of the true particle track. For the cases in which a coordinate cannot be determined (applicable to x- and y-coordinates), the reconstructed coordinate is set to the center of the active volume. The vertex error is then defined as the difference between the reconstructed and true coordinates.

Given that the distributions are not Gaussian, we can still attempt to fit a Gaussian function to the vertex error distributions by ignoring the tails to extract a vertex resolution from the FWHM of the fit as a figure of merit. This is done by fitting a Gaussian function to the core of the distribution.

The results of this fitting procedure as a function of true neutrino energy are presented in fig. 33 for both the neutrino-only and neutrino-plus-background samples.

Table 9 summarizes the vertex resolution for all geometry and background configurations. The vertex resolution is found to be between 0.5 cm and 0.75 cm. The fact that all coordinates and configurations result in a similar vertex resolution points to the fact that the vertex resolution is dominated by the intrinsic smearing of the point-like reconstruction approach, which places the reconstructed vertex at the center of the neutrino-induced particle track. As the neutrino energy increases, the average particle track size also increases, leading to a larger displacement between the true interaction vertex and the reconstructed vertex. Currently, there is no low–energy physics analysis in DUNE that would strongly benefit from a vertex resolution better than a few

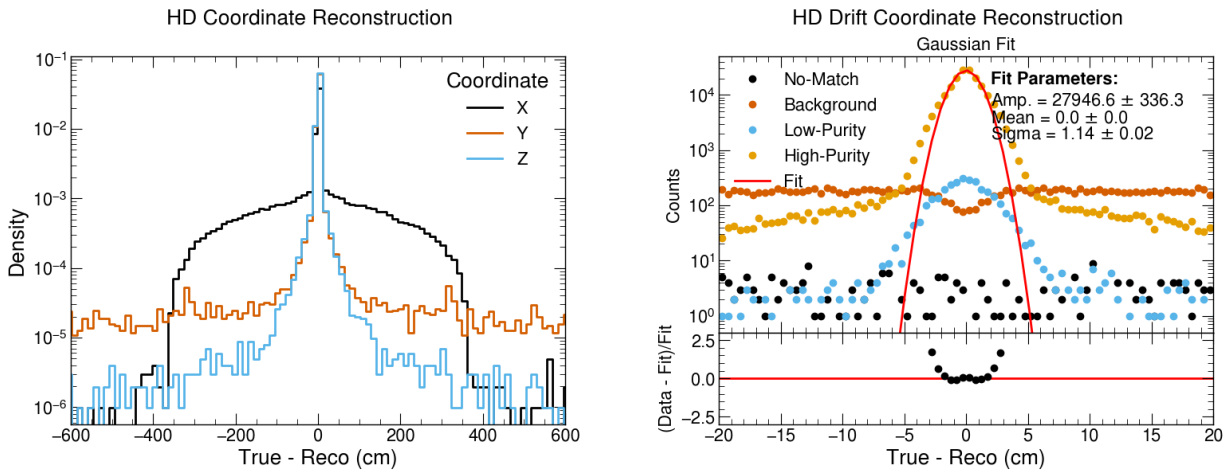


Figure 33: HD vertex error distributions. (left) All coordinate error distributions. (right) X coordinate error divided in high- and low-purity matched flash samples with core distribution fit.

Vertex Resolution (cm))			
Configuration	Vertex		
	X	Y	Z
HD Signal	0.72 ± 0.01	0.55 ± 0.01	0.64 ± 0.01
HD Lateral	0.65 ± 0.03	0.68 ± 0.01	0.65 ± 0.01
HD Central	0.71 ± 0.02	0.65 ± 0.01	0.65 ± 0.01
VD Signal	0.69 ± 0.01	0.69 ± 0.01	0.67 ± 0.01
VD Top	0.58 ± 0.03	0.66 ± 0.01	0.68 ± 0.02

Table 9: Vertex resolution comparison between all geometry and background configurations. The values are extracted from the FWHM of the Gaussian fit to the core of the vertex error distribution.

centimeters. For example, in the case of solar neutrinos, where external cavern–gammas are one of the major backgrounds, the Compton scattering length is on the order of tens of centimeters, which is much larger than the achieved vertex resolution. But, if needed, the vertex resolution could be improved by implementing a more sophisticated reconstruction algorithm that takes into account the topology of the event.

Figure 34 shows the vertex reconstruction efficiency as a function of true drift coordinate for different neutrino energies. The efficiency is defined as the fraction of events for which the reconstructed vertex is within a certain distance (3 widths of the core error distribution) from the true vertex position. As expected, the efficiency increases with increasing neutrino energy, as higher energy events tend to produce more ionization charge and scintillation light, leading to better vertex reconstruction accuracy.

5.3 Fiducialization Efficiency

In order to reduce the impact of background events originating from the detector materials and the surrounding environment, a fiducial volume cut is typically applied to the reconstructed neutrino events in low energy-analyses. This cut is based on the reconstructed position of the

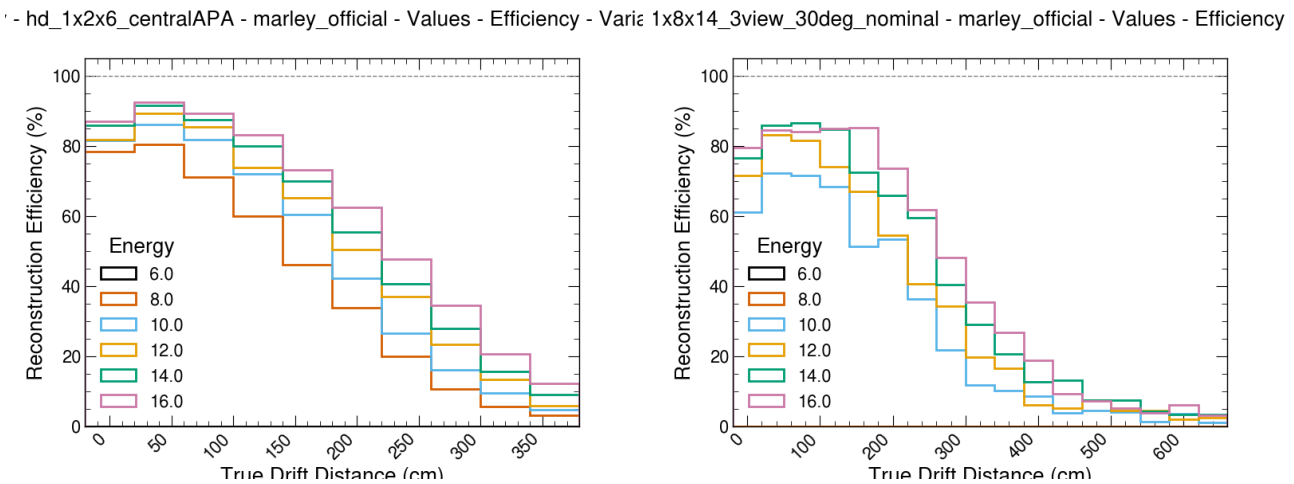


Figure 34: HD (left) and VD (right) vertex reconstruction efficiency as a function of true neutrino energy for different vertex error tolerances.

primary TPC cluster associated with the neutrino interaction. The fiducial volume is defined as a region within the detector that is at least x cm away from any external detector boundary. This distance can be chosen in any analysis to optimize the signal-to-background ratio.

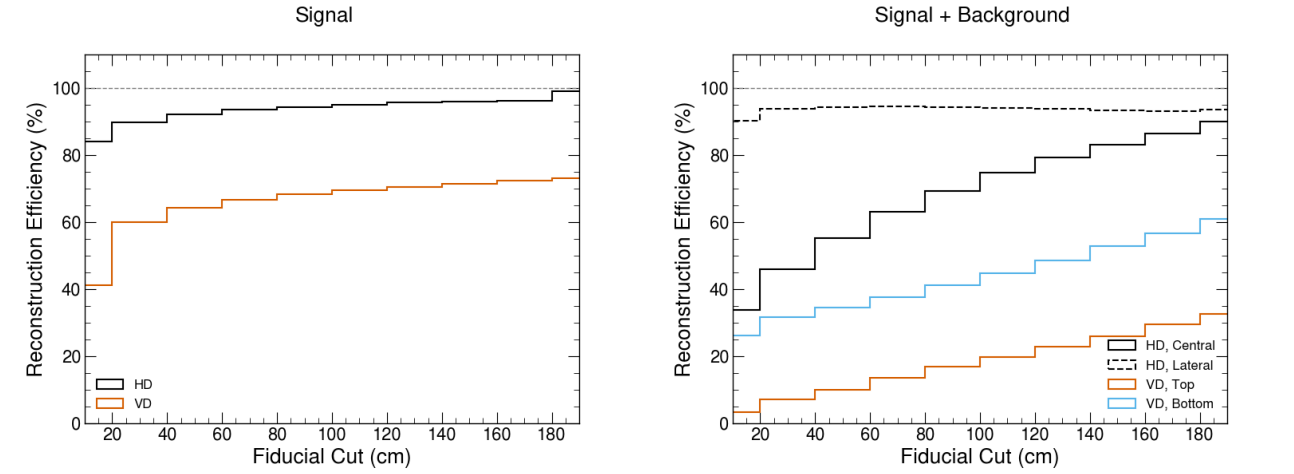


Figure 35: Fiducialisation efficiency as a function of cut-distance to the detector boundary for the average neutrino energy sample of 4 MeV to 30 MeV.

The results for the HD geometry with central APA background scenario are shown in fig. 35 as a function of the distance to the detector boundary for different neutrino energies. The efficiency is calculated as the fraction of events with a reconstructed vertex position that lies within the fiducial volume, relative to the total number of simulated neutrino interactions occurring within the fiducial volume. As expected, the efficiency increases with increasing distance from the detector boundary, as events occurring closer to the boundary are more likely to be miss-reconstructed. Additionally, the efficiency improves with increasing neutrino energy, as higher energy events tend to produce more ionization charge and scintillation light, leading to better vertex reconstruction accuracy.

For a better comparison between the modules, we evaluate the average fiducialisation efficiency up to 100 cm for 10 MeV neutrino energies. The results for the HD and VD geometries are shown in table 10, respectively. The efficiency values are generally high, especially in the y- and z-coordinates, where they exceed 65% in all cases. The x–coordinate shows a lower efficiency, particularly for the VD geometry, which can be attributed to the larger drift volume and lower

Fiducialisation Efficiency (%) – 10 MeV			
Configuration	Coordinate		
	X	Y	Z
HD Signal	71.6 ± 0.7	65.7 ± 0.8	89 ± 3
HD Lateral	78 ± 1	65 ± 2	80 ± 2
HD Central	18.6 ± 0.6	65 ± 2	98 ± 9
VD Signal	36.1 ± 0.6	78.9 ± 1.0	76 ± 1
VD Nominal	4.5 ± 0.8	73 ± 4	78 ± 6

Table 10: Average vertex fiducialisation efficiency at 100 cm from the detector boundary.

TPC–PDS matching efficiency. The central HD volumes also have a lower fiducialisation capacity towards their edges, but thanks to the intrinsic fiducialisation provided by the lateral APAs.

5.4 Electron Energy Resolution

As a figure of merit for the process, we can compute the energy resolution from the calibrated energy of the main electron. To remove any reconstruction artifacts, this first result uses:

- **Ideal Clustering**. This means that all the hits associated with the electron are perfectly clustered together, and no hits from other particles are included. This is done using truth information to select the hits.
- **High Hit Threshold**. A minimum of 3 hits is required to form a cluster. This helps to reduce noise and ensures that only significant energy depositions are considered.
- **True Drift Correction**. The drift correction is applied using the true position of the interaction, rather than relying on reconstructed positions. This eliminates any errors that might arise from inaccuracies in the reconstruction process.- Three scenarios are considered: **True** drift correction, **Reco** drift correction (using flash-matching), and **Mean** drift correction (using the average drift time for the detector).

This is typically the value that gets compute when estimating the energy resolution of a TPC. It is obtained for an idealized clustering with ad hoc calorimetry and a high hit threshold (in this case we use 3 for a typical track). The energy resolution is computed as the Root Mean Square (RMS) of the relative error of the reconstructed variable versus the truth. The results are shown in **??**. The formula for the energy resolution fit is:

$$\Omega_{(E_{\text{true}})} = \sqrt{\left(\frac{p_2}{\sqrt{E_{\text{true}} - b}}\right)^2 + \left(\frac{p_1}{E_{\text{true}} - b}\right)^2 + p_0^2}, \tag{17}$$

where p_0, p_1, and p_2 are the parameters of the fit, and b is a constant offset and corresponds to the detection threshold. These parameters can be interpreted as the intrinsic resolution evaluated to 11(1)%, the stochastic term, and the noise term, respectively. This formula con be derived from the standard parameterization of energy resolution in calorimetry, which typically includes contributions from stochastic processes, electronic noise, and constant terms.

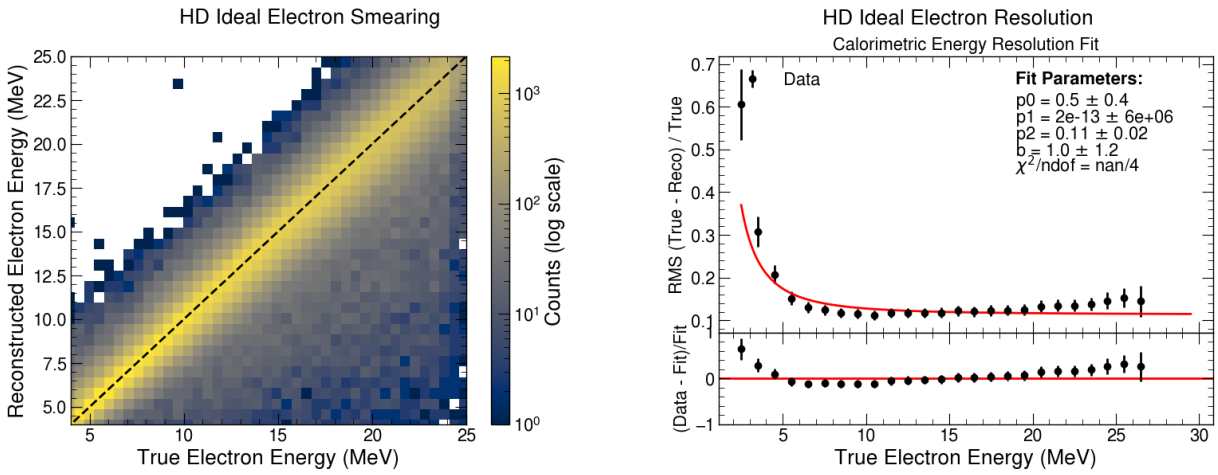


Figure 36: Energy resolution from calibrated charge with ideal clustering, true drift correction and hit threshold of 3. (Left) Heatmap of the reconstructed energies. (Right) Per bin resolution of the data.

To assess the impact of the different reconstruction stages on the energy resolution, we can vary one parameter at a time while keeping the others fixed. The results are shown in fig. 37.

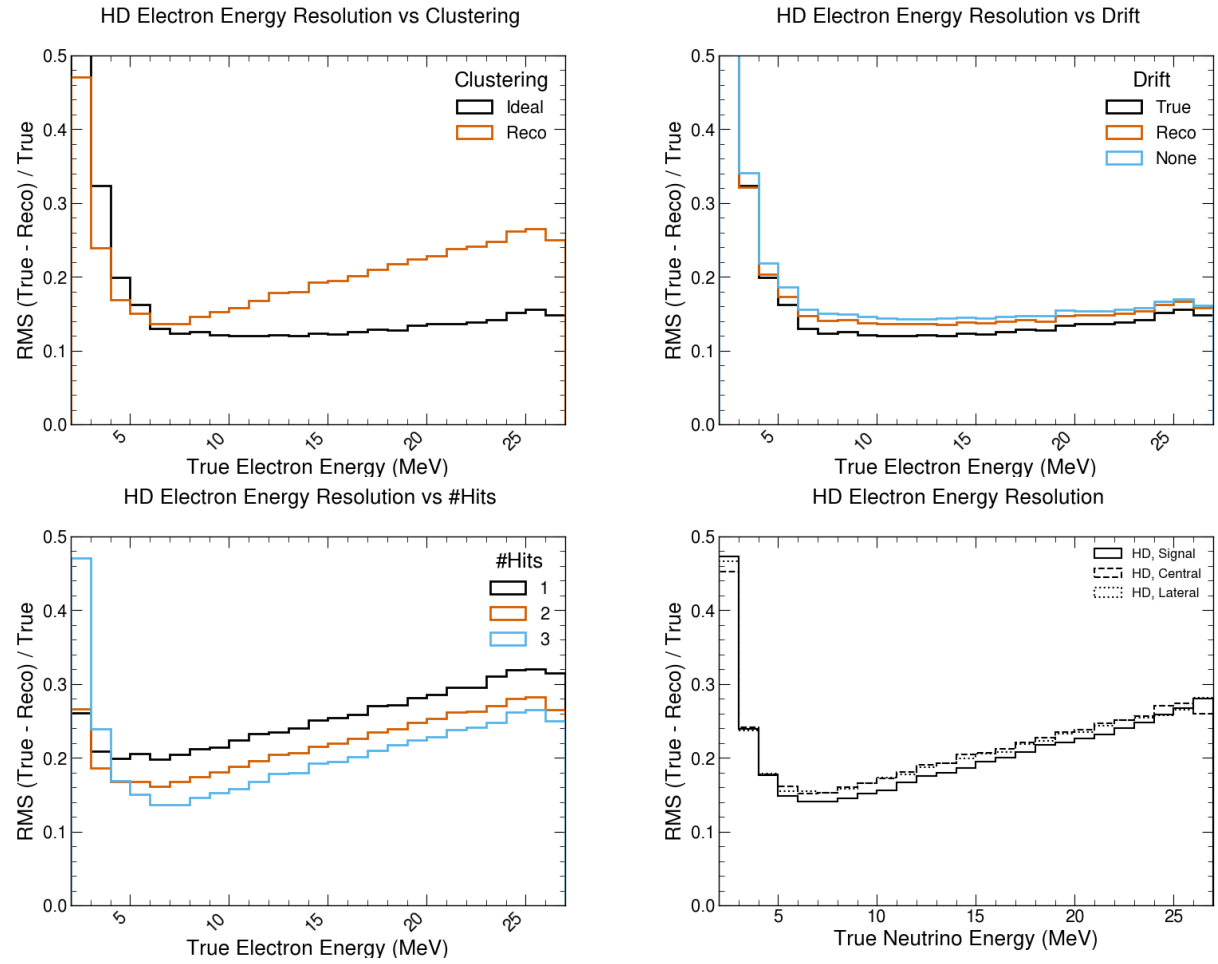


Figure 37: Effect of clustering (ideal vs reconstructed), drift correction (ideal vs reconstructed vs none), and hit threshold on the electron energy resolution. (Top Left) Shows the effect of non-ideal clustering. (Top Right) Shows the effect of not applying a drift correction. (Bottom Left) Shows the effect of lower hit threshold. (Bottom Right) Shows the effect of background.

As can be seen, the clustering has the most significant impact on the energy resolution, especially with increasing energies. This is expected, as the clustering algorithm may not be able to perfectly reconstruct the electron, leading to a loss of energy. The drift correction has a small but noticeable impact on the energy resolution. This is accentuated in the case of the VD geometry (see fig. 41). The hit threshold also has an impact on the energy resolution, but it is less significant than the clustering. A higher hit threshold leads to a better energy resolution, as it selects electrons with higher completeness. Finally, the presence of background has a significant impact on the energy resolution, especially for the VD geometry, where the energy resolution degrades by a factor of 2 in the 4 MeV to 30 MeV range. This is expected, as the background can lead to mis-reconstruction of the electron and therefore a loss of energy. The effect of background is less pronounced in the HD geometry due to its smaller drift volume and lower background rate.

5.5 Neutrino Energy Resolution

The energy resolution is calculated from the calibrated neutrino energies as the RMS of the relative error of the reconstructed variable versus the truth. The results are shown in fig. 38 for the 4 different reconstruction algorithms. The energy resolution is evaluated as a function of the true neutrino energy, which is obtained from the simulation truth information. This allows us to assess how well the reconstructed energy matches the true energy across a range of neutrino energies, providing insight into the performance of the reconstruction algorithms.

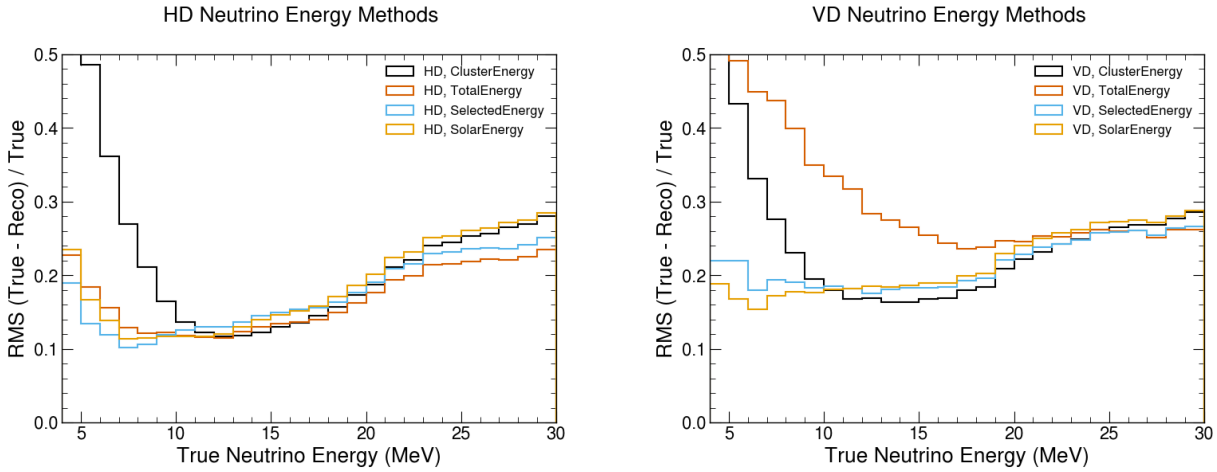


Figure 38: Neutrino energy resolution for the different reconstruction algorithms. (left) HD geometry. (right) VD geometry.

As expected, the *ClusterEnergy* algorithms has the worst energy resolution for neutrino energies below 10 MeV, as it does not take into account the energy deposited by adjacent gamma blips which can amount to more than 50% of the initial neutrino energy in this range. The *TotalEnergy* algorithm, on the other hand, has a good energy resolution in the full range for HD, but it is significantly worse for the VD geometry due to the higher background rate and lower clustering efficiency.

The *SelectedEnergy* and *SolarEnergy* algorithms provide a significant improvement in the energy resolution for neutrino energies below 30 MeV by maximizing the adjacent cluster significance. The advantages of the *SolarEnergy* algorithm becomes apparent for event candidate reconstruction in the context of solar neutrino analyses, where the signal energy range is expected to be

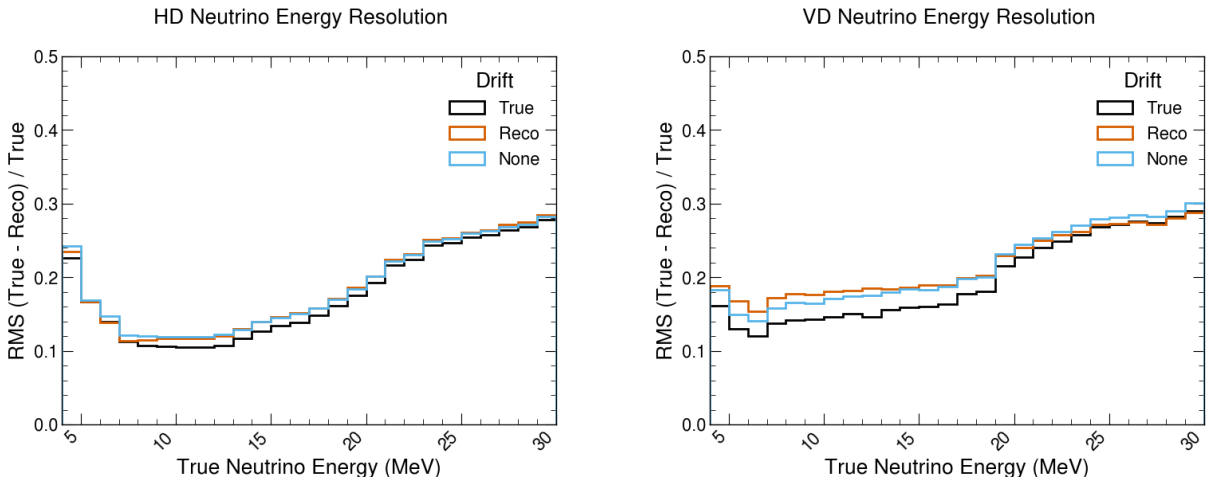


Figure 39: (left) HD (right) VD. Neutrino energy resolution with and without flash-matching. The labels **True**, **Reco** and **None** represent the cases for which the drift coordinate is taken from truth information, from the flash-matching step and with an average drift correction, respectively.

dominated by external gammas and neutrons from the cavern. The latter produce a gamma cascade which results in an enhanced adjacent cluster activity. If not limited algorithmically (e.g. by applying a maximum adjacent cluster energy as in the *SolarEnergy* method), this can lead to a significant overestimation of the event's energy. Further details on this will be provided in the solar neutrino analysis note.

Figure 39 shows the effect of the flash-matching step on the energy resolution. The flash-matching step allows to associate a TPC cluster with the corresponding PDS flash and hence obtain the correct drift coordinate of the interaction. The difference between the flash-matched and non-flash-matched cases are small for HD, indicating that the inefficiencies in the energy reconstruction are dominated by other factors, such as clustering inefficiencies and intrinsic energy losses due to nuclear effects and undetected particles. In the case of VD, the flash-matching step has a more noticeable effect on the energy resolution ($\sim$5 % in the 4 MeV to 20 MeV range). This is expected due to the larger drift volume.

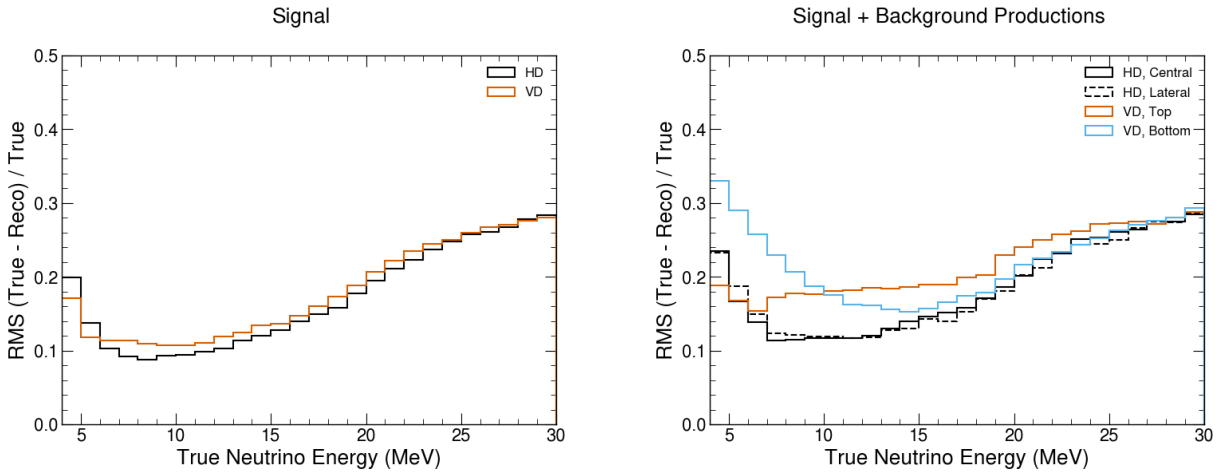


Figure 40: Energy resolution comparison between all geometry and background configurations. (left) Comparison between the hd_1x2x6_centralAPA and vd_1x8x14_3view_30deg geometries. (right) Comparison between the different background scenarios for the vd_1x8x14_3view_30deg geometry.

Finnaly, fig. 40 shows the energy resolution comparison between all geometry and background configurations. We can see that the energy resolution is similar between the hd_1x2x6 and vd_1x8x14_3view_30deg geometries, with a noticeable advantage for the former in the configurations with background (hd_1x2x6_centralAPA & hd_1x2x6_lateralAPA vs vd_1x8x14_3view_30deg_nominal & vd_1x8x14_3view_30deg_shielded).

6 Conclusion

In this technical note, we have presented the reconstruction algorithms for the DUNE FD, focusing on the neutrino energy and vertex resolution. The reconstruction is based on the TPC cluster vertex reconstruction and flash–matching, which allows us to extract the calibrated neutrino energy and vertex position.

The vertex and energy reconstruction of low–energy neutrino interactions in DUNE has been studied. The vertex resolution was evaluated using a point–like reconstruction approach that combines the TPC cluster vertex with the associated PDS flash. The achieved vertex resolution is between 0.5 cm and 3.5 cm, depending on the coordinate and the presence of background events. There are significant differences in fiducialisation performance between the HD and VD geometries, with the HD geometry achieving higher fiducialisation efficiency due to its lower background rates. The energy reconstruction was performed using four different algorithms to sum the energy of reconstructed TPC clusters. The performance of these algorithms was evaluated in terms of energy resolution, fiducialisation efficiency, and the impact of flash–matching on the energy reconstruction. Contrary to the vertex resolution, the energy resolution does not show significant differences between the two geometries provided that one chooses the appropriate reconstruction algorithm. The best energy resolution is achieved using the *SolarEnergy* algorithm, which yields a resolution of approximately 20% at 15 MeV and improves to about 10% at 5 MeV. The flash–matching step has a minor impact on the energy resolution, indicating that other factors, such as clustering inefficiencies and intrinsic energy losses, dominate the resolution. Overall, the reconstruction algorithms demonstrate good performance for low–energy neutrino interactions in DUNE, with potential for further optimization and improvement. Based on the amount of background activity in the VD module, other reconstruction objectives such as channel tagging (out of the scope of this work) and particle identification are likely to be more challenging than in the HD module. The case of a shielded VD module, which has background rates comparable to the HD module, is therefore highly encouraged and expected to yield similar low–energy reconstruction performance.

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Acronyms

APA Anode Plane Assembly. 4, 5, 6, 7, 28, 40, 41

CC Charged Current. 1, 3, 6, 8, 9, 10, 16, 25, 26, 28, 29, 35

CPA Cathode Plane Assembly. 4, 5, 6, 7, 28

CRP Charge Readout Plane. 4, 5, 7

ES Elastic Scattering. 8, 10, 25, 26

FD Far Detector. 1, 2, 3, 4, 5, 6, 12, 25, 45

Fermilab Fermi National Accelerator Laboratory. 4

Geant4 Geometry and Tracking v.4. 6, 8, 12, 13

LAr Liquid Argon. 4, 5, 6, 8, 9, 10, 11, 12, 13, 16, 17, 18, 21, 22, 24, 25, 30

LArSoft Liquid Argon Software Framework. 6, 8, 11, 15

LArTPC Liquid Argon Time Projection Chamber. 2, 3, 4, 5, 12, 13

MC Monte Carlo. 6, 8, 9, 15

NC Neutral Current. 8

PCB Printed Circuit Board. 4, 5

PD Photon Detector. 14

PDE Photon Detection Efficiency. 14

PDS Photon Detection System. 2, 3, 5, 14, 15, 18, 19, 20, 21, 22, 24, 29, 37, 44

PE Photoelectron. 14, 18, 19, 22, 23, 24

RMS Root Mean Square. 41, 43

ROI Region Of Interest. 14

SiPM Silicon Photomultiplier. 14

SURF Sanford Underground Research Facility. 4, 10

TPC Time Projection Chamber. 2, 3, 5, 6, 7, 11, 14, 15, 18, 20, 21, 22, 23, 24, 25, 26, 28, 29, 37, 40, 41, 44, 45

VUV Vacuum Ultraviolet. 12

XA X-ARAPUCA. 5, 14, 18, 19

Glossary

Decay0 is a program written in the early 90's by Volodymyr Tretyak and collaborators, using the Fortran 77 programming language and the CERNLIB library. It has been maintained and improved up to 2020. Later, a pure C++ version was released without any dependency on Fortran and CERNLIB. This code was integrated in 2011 as the genbb_help module of the Bayeux C++ library (version ¡ 4). BxDecay0 is a standalone library with very few dependencies (mostly the GSL library for numerical integration and a few special math functions). External random engines can be used through a simple wrapping functor interface, particularly the ones provided by the C++ standard random library can be used by default. However the user is free to provide its own uniform deviates random generator (based on GSL, ROOT or whatever).. 6, 8

DUNE (Deep Underground Neutrino Experiment) is a next-generation neutrino experiment designed to study neutrino oscillations, proton decay, and supernova neutrinos. It will consist of a far detector located deep underground at the Sanford Underground Research Facility (SURF) in South Dakota, and a near detector located at Fermilab in Illinois. DUNE aims to provide insights into the fundamental properties of neutrinos and their role in the universe.. 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 15, 25, 26, 45, 52

HD (Horizontal-Drift) is a design for a liquid argon time projection chamber (LArTPC) where the electric field is oriented horizontally. In this design, the anode plane assemblies (APAs) are arranged vertically, and the cathode plane assemblies (CPAs) are placed horizontally. This configuration allows for a shorter drift distance, which can lead to improved charge collection and better spatial resolution. The HD design is one of the two geometries planned for the DUNE far detector modules.. 4, 5, 6, 7, 14, 15, 19, 20, 21, 23, 24, 25, 26, 27, 28, 30, 31, 32, 33, 37, 38, 39, 40, 41, 43, 44, 45

LArQL is a software package developed to simulate the response of liquid argon time projection chambers (LArTPCs). It provides tools for modeling the scintillation processes in liquid argon. LArQL is designed to be used in conjunction with the LArSoft framework for simulating and reconstructing events in LArTPC detectors [**?**].. 8, 12

MARLEY (Model of Argon Reaction Low Energy Yields) is a neutrino event generator designed to simulate low-energy neutrino interactions with argon nuclei. It is developed by the Low-Energy Neutrino group of the DUNE collaboration. MARLEY focuses on simulating interactions relevant to solar and supernova neutrinos in liquid argon detectors, providing detailed modeling of the nuclear physics involved in these interactions. It incorporates a gamma-decay scheme to describe the de-excitation of the residual nucleus after a neutrino interaction, which is crucial for accurately simulating the resulting signals in the detector. MARLEY is implemented in C++ and is designed to be integrated with the LArSoft framework used by the DUNE collaboration for detector simulation and event reconstruction.. 6, 8, 9, 12, 26

MicroBooNE is a neutrino experiment located at Fermilab in Illinois. It is designed to study neutrino interactions using a liquid argon time projection chamber (LArTPC) detector. MicroBooNE is part of the Short-Baseline Neutrino (SBN) program, which aims to investigate the nature of neutrino oscillations and search for sterile neutrinos. The experiment has been operational since 2015 and has provided valuable data on neutrino interactions in liquid argon.. 14

ProtoDUNE-SP is a prototype detector for the DUNE experiment, located at CERN in Switzerland. It is a single-phase liquid argon time projection chamber (LArTPC) designed to test and validate the technologies and techniques that will be used in the DUNE far detector. ProtoDUNE-SP has been operational since 2018 and has provided valuable data on the performance of LArTPC technology, as well as insights into neutrino interactions in liquid argon.. 14

VD (Vertical-Drift) is a design for a liquid argon time projection chamber (LArTPC) where the electric field is oriented vertically. In this design, the anode plane assemblies (APAs) are arranged horizontally, and the cathode plane assemblies (CPAs) are placed vertically. This configuration allows for a longer drift distance, which can lead to improved energy resolution and better background rejection. The VD design is one of the two geometries planned for the DUNE far detector modules.. 4, 5, 7, 14, 19, 20, 21, 22, 23, 24, 25, 26, 27, 30, 31, 32, 40, 43, 44, 45

WireCell is a software package developed to simulate the response of liquid argon time projection chambers (LArTPCs). It provides tools for modeling the ionization processes in liquid argon, as well as the electronics response of the detector. WireCell is designed to be used in conjunction with the LArSoft framework for simulating and reconstructing events in LArTPC detectors [12].. 8, 13, 14, 51, 52

7 Appendix

Introduction

Due to its nature as a noble gas, ^{222}Rn is difficult to remove from the detector environment, making it a persistent background source also in detector components. Among its progeny, several isotopes are particularly important:

- **Bismuth (^{214}Bi)**: A short–lived beta emitter with a high Q–value (up to 3.27 MeV) and multiple gamma lines. It is the most significant source of high–energy background from ^{222}Rn [15].
- **Bismuth (^{210}Bi)**: A beta emitter that produces a continuum low–energy background up to 1.16 MeV.
- **Lead (^{210}Pb)**: A long–lived isotope ($t_{1/2}$ = 22.3 years) that accumulates on detector surfaces [16].
- **Polonium (^{210}Po)**: An alpha emitter with a half–life of 138 days. It can be used as a surface contamination proxy due to its distinct alpha signal [17].

TPC Signal Processing with Wire-Cell

WireCell is a software toolkit for signal processing and reconstruction in liquid argon TPCs. It provides a comprehensive framework for converting raw digitized waveforms into high-level physics objects, such as 3D space points, clusters, and tracks. The key features of WireCell include:

- TPC Signal Processing (2D Deconvolution).The initial stage involves converting raw digitized waveforms into the distribution of ionization electrons.
  - **Noise Filtering**: Excess noise is statistically removed from the digitized TPC waveforms.
  - **2D Deconvolution**: This technique accounts for the detector response in both the *time* and *wire* dimensions. It statistically unfolds the electronics response and the long-range induction field response.
- Tomographic 3D Imaging. WireCell utilizes tomographic principles to resolve the inherent ambiguities of wire-plane readouts.
  - **Slicing and Tiling**: Deconvolved signals are divided into time slices (typically ∼2–3 μs), and wires with signals are overlapped to identify potential 3D "cells" or "blobs".
  - **Solving for Ghosts**: To remove "fake" hits (ghosts) created by the wire geometry, Wire-Cell applies **compressed sensing** (L1 regularization) to solve a non-deterministic linear system.

- **Mathematical Objective**: The software minimizes a χ^2 function with a regularization term:
$$\chi^2 = \|y' - A'x\|_2^2 + \lambda\|x\|_1 \tag{18}$$
where x represents the true charge to be resolved and y represents the measured wire signals.

- 3D Clustering and Charge-Light Matching. Reconstructed 3D space points are grouped into clusters based on connectivity.
  - **Event Timing (t_0)**: Because TPC charge signals and scintillation light are decoupled, Wire-Cell performs many-to-many matching between TPC clusters and PMT flashes.
  - **Matching Principle**: This process utilizes the proportionality of the light signal to the reconstructed 3D charge to determine the precise time of the neutrino interaction.

- 3D Trajectory and dQ/dx Fitting. Once t_0 is determined, high-resolution tracking is performed.
  - **Trajectory Fit**: A fine-grained 3D path is determined for each cluster using algorithms such as the Steiner-tree-inspired graph approach [19-21].
  - **Calorimetry (dQ/dx)**: Ionization charge per unit length (dQ/dx) is fitted along the track. This information is critical for Particle Identification (PID), allowing the software to distinguish between particles like protons, muons, and electrons [2, 23, 24].

- A key features in WireCell is its **topology-agnostic** approach, making it effective for both simple tracks and complex electromagnetic showers. It significantly reduces the impact of non-functional wires by allowing reconstruction with only two functional planes, reducing dead volume from ∼30% to ∼3%.

Currently in DUNE it is used to model the drift of ionisation electrons towards the TPC wire planes, accounting for diffusion, recombination, and attachment effects. The resulting charge distribution at the wire planes is then converted into the induced signals on the wires. In future simulations, the WireCell toolkit could be used to provide all of its reconstructions goals.

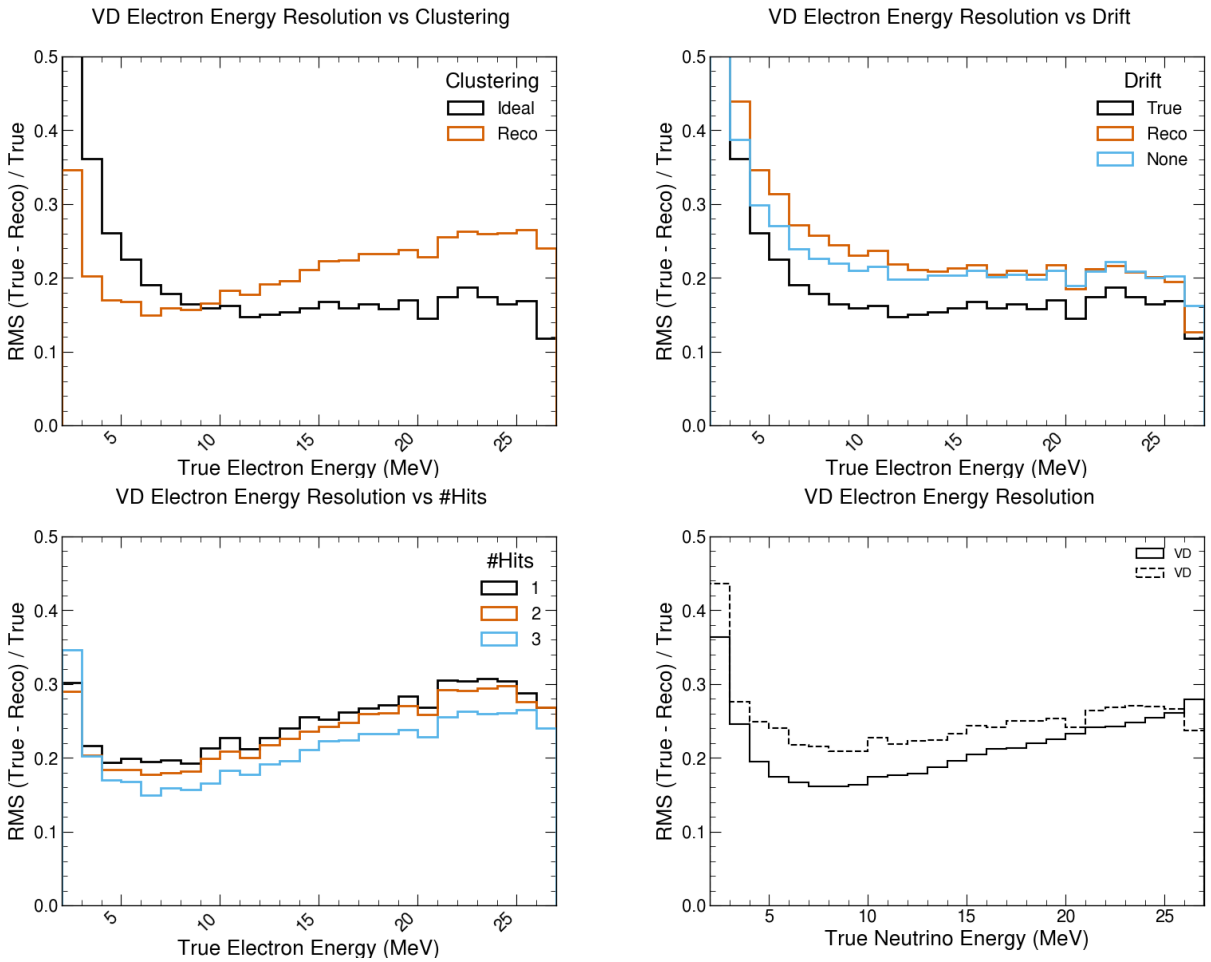


Figure 41: Effect of clustering (ideal vs reconstructed), drift correction (ideal vs reconstructed vs none), and hit threshold on the electron energy resolution. (Top Left) Shows the effect of non-ideal clustering. (Top Right) Shows the effect of not applying a drift correction. (Bottom Left) Shows the effect of lower hit threshold. (Bottom Right) Shows the effect of background.